Monopoly, Product Quality, and Flexible Learning *

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Abstract

A seller offers a buyer a schedule of transfers and associated product qualities. After observing this schedule, the buyer chooses a flexible costly signal about his type. We show it is without loss to focus on a class of allocations that compensate the buyer for his learning costs. We show that with strictly increasing marginal quality costs, quality lies strictly below the efficient level, even "at the top." Moreover, with constant marginal quality costs, the optimal menus have intermediate options. We also provide sufficient conditions for the optimal menu to be simple.

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1. Introduction

The technological advancements of the last few decades have made it easier for consumers to learn about products before trading. When choosing what information to acquire, buyers commonly rely on the set of available products and trade terms. Consider a consumer shopping for a mobile-phone subscription, for example. Such a consumer would have to obtain a finer estimate of his expected phone usage to evaluate a pay-per-minute plan than he would for a plan with unlimited calls. Because the buyer's willingness to pay depends on her information, the seller will likely consider the impact her menu has on the buyer's learning decisions when choosing what contracts to offer. For instance, adding novel features to one's products may be pointless if consumers never invest in learning about these features before purchasing. In this paper, we study how the need to guide the buyer's learning influences the menu offered by a monopolist with vertically differentiated products.

Specifically, we study a model in which a seller of vertically differentiated products decides what menu to offer to a potential buyer. Unlike the classical model of Mussa and Rosen (1978) and Maskin and Riley (1984), we do not assume the buyer possesses private information when he first sees the monopolist's menu. Instead, the buyer sees this menu, and then *chooses* what to learn about his type. The buyer's information choice is flexible and costly; we expand on these assumptions below. The monopolist's menu designates a schedule of qualities and associated transfers, where the monopolist's marginal costs can be either constant or strictly increasing with quality. Our main interest is in the structure of this menu and the efficiency of the resulting allocation with respect to the buyer's chosen information.

The key message of our analysis is that accounting for the interaction between the monopolist's offerings and buyer learning changes the parameters that determine the monopolist's optimal menu. When the buyer's information is fixed, the buyer's information rents can only reduce the monopolist's profits. By contrast, in our setting the monopolist uses these rents to incentivize the buyer to collect the right information. Consequently, the monopolist's optimal product-line is no longer a function only of the buyer's information (as is the case in Mussa and Rosen, 1978; Maskin and Riley, 1984), but is also affected by the structure of the buyer's costs of learning. We show this additional effect has concrete implication: it creates additional inefficiencies, and yields new explanations for the use of simple menus.

We now describe the buyer's preferences, signal choice, and cost of information. We assume the same buyer preferences as in Mussa and Rosen (1978). Specifically, we postulate the buyer's preferences are quasi-linear in money, and that his marginal utility from quality is constant and equal to his type, $\theta \in \Theta = \left[\underline{\theta}, \bar{\theta}\right]$. Combined with expected-payoff maximization, this preference specification implies that the mean of the buyer's posterior belief pins down his payoffs from any quality-transfer pair, and through it, his selection from any menu. Consequently, for any fixed menu, the distribution of the buyer's posterior mean fully determines trade outcomes. Thus, we summarize every information structure the buyer can choose by the distribution of posterior estimates it induces. The buyer's learning choice is fully flexible, meaning the buyer can choose any distribution for her posterior type estimate that is consistent with some signal structure. Following Ravid, Roesler, and Szentes (2022), we define the cost of information acquisition directly as a function of this distribution. In particular, we postulate that this function is affine and increasing in informativeness, which we show is equivalent to the cost of each distribution being equal to its integral against a convex function, c.[1] Mensch and Malik (2023) refers to such cost functions as *posterior-mean separable*, and characterizes their revealed preference properties.

Our modeling assumptions imply the buyer's optimal learning program is a special case of the more general mean-measurable information design problem (Gentzkow and Kamenica, 2016; Dworczak and Martini, 2019; Arieli et al., 2020; Kleiner, Moldovanu, and Strack, 2021). Specifically, the buyer chooses a cumulative distribution function (CDF) for his posterior type estimate in order to maximize the integral of some function. In our case, this function equals the buyer's net utility, which is his payoff from choosing the optimal option from the monopolist's menu given his realized posterior estimate (i.e., type) θ, minus $c(\theta)$. The buyer is constrained to choosing CDFs that arise from some signal, which is equivalent to choosing a CDF from which one can attain the true type distribution via mean-preserving spreads.

Unlike their more general counterpart, mean-measurable information design problems remain tractable even when the underlying state space is very large. Consequently, we can accommodate discrete and continuous prior distributions. Moreover, using Dworczak and Martini's (2019) duality-based tools, we can show it is without loss to focus on the class *information-cost-canceling* (ICC) allocations. These allocations decompose the buyer's information rents into two parts: one part that cancels out the buyer's costs of learning, and a residual that comes from the derivative of the "price function" from Dworczak and Martini's (2019) duality characterization.

The reduction to ICC allocations enables us to reason about the monopolist optimal outcome using two familiar-looking programs. The first program involves maximizing the monopolist's profits across all cost canceling allocations that make a fixed distribution incentive compatible. The reduction to ICC allocations makes this program similar to the standard mechanism design programs where the "choice variable" is the set of all bounded increasing functions. The second program searches for the most profitable signal for the monopolist among those signals that are incentive compatible for the buyer given a fixed allocation. Whenever this allocation is ICC, the second program turns out to be a "constrained" mean-measurable information design problem.

Using these two programs, we prove several results about the monopolist optimal outcome.

[1] This assumption appears in the leading example of Ravid, Roesler, and Szentes (2022).

Our first main result shows that when marginal costs of quality are increasing and learning costs are sufficiently steep near $\bar{\theta}$, the buyer's chosen quality always lies strictly below the efficient level conditional on his signal realization. This strict downward distortion of quality holds even when the buyer's posterior type is the highest possible given his signal, a feature that stands in contrast to the case in which the buyer's information is fixed. In that case, it is well known that the monopolist's optimal allocation involves "no distortion at the top": the type with the highest value in the distribution receives the efficient quality level.

Our second result considers the case where the marginal costs of quality are constant. In this case, the efficient outcome has all types consuming the maximal possible quality to all types. Under exogenous information, it is well-known the monopolist optimal allocation involves serving this maximal quality to all types above some threshold, where this threshold is typically interior (e.g., Riley and Zeckhauser, 1983). All types below this threshold are excluded. By contrast, we show that, in our model, the monopolist serves the maximal quality only to the buyer with the highest possible signal realization. All other signal realizations get a strictly lower quality. Moreover, only the lowest possible realized type is ever excluded.

We also derive some results about the buyer's information structure at the monopolist optimal outcome. In particular, we show it is without loss to restrict attention for the buyer's signal to be a bipooling (Arieli et al., 2023). Such signals partition the type space into intervals of two types. The first type of intervals are ones where the type is revealed. The second type are "bipooling" intervals, where all types are pooled into one of two signal realizations within the interval. We also obtain conditions under which one can further restrict the buyer to binary signals. Specifically, we show such signals are without loss of optimality if learning costs have a steep slope, or if the seller's marginal costs for quality are constant and the seller's profit satisfies a concavity condition.

The structure of the buyer-optimal signal allows us to make inferences about the size of the monopolist optimal menu. In particular, whenever the buyer's signal is binary, a revelation-principle style argument immediately delivers the monopolist can maximize profits by offering the buyer at most two purchasing options. In fact, such circumstances turn out to be the *only* ones in which the monopolist can attain her optimum using a two-option menu. Whenever the buyer uses a richer signal, the monopolist-optimal menu must include more alternatives. On the flip side, there are circumstances where the buyer's signal is binary, but the seller can attain the optimum using a menu with a single purchasing option. We identify such sufficient conditions in Corollary 4.

Related Literature. Our paper lies in the literature studying the interaction between flexible information acquisition and trade. Very closely related is Mensch (2022), who studies the optimal way to auction an indivisible good with zero production costs to buyers who flexibly acquire information about their value after observing the menu. While our constant marginal cost case is reminiscent of this environment, especially with binary states, our more general model, using posterior-mean separable information costs, yields new insights coming from the duality-based characterization of ICC mechanisms that his model cannot deliver with merely posterior-separable costs. Moreover, while his results yield efficency (i.e. sale with probability 1) at the highest signal, in contrast we find inefficiently low quality provision at all signals, including the highest, whenever marginal costs are strictly increasing.

In a concurrent paper, Thereze (2023) analyzes a variant of our model in which the monopolist's marginal costs for quality are strictly increasing and the buyer's type is binary, assuming information costs are posterior separable. Using the techniques of Mensch (2022), he finds, like us, that there is downward distortion of quality for all types, including at the top. He also shows that the seller's profits are not monotone in the buyer's costs of information. By assuming posterior-mean separable costs, our approach allows us to accommodate multiple/continuum of states, and obtain results about the shape of the optimal signal and the size of the monopolist's menu that do not appear in Thereze (2023).

In addition to Mensch (2022) and Thereze (2023), several other papers study the interaction between information design (Aumann and Maschler, 1995; Kamenica and Gentzkow, 2011; Bergemann and Morris, 2013) and trade. The closest papers to ours are Condorelli and Szentes (2020) and Ravid, Roesler, and Szentes (2022), both of which study models of bilateral trade with a single indivisible good, but differ in the timing of information acquisition. Several other papers study the set of possible outcomes in bilateral trade settings with indivisible goods as one varies each party's information; see Bergemann, Brooks, and Morris (2015); Roesler and Szentes (2017); Kartik and Zhong (2019); Haghpanah and Siegel (2022a);and Haghpanah and Siegel, 2022b.[2]

More broadly, this paper contributes to the burgeoning literature on rational inattention, started by the seminal papers of Sims (1998; 2003), and developed into models of flexible information acquisition by Caplin and Dean (2013; 2015), Matějka and McKay (2015), and Caplin, Dean, and Leahy (2021) using a posterior-separable approach to modeling information costs. Since then, there have been a number of applications of rational inattention to various economic problems, such as global games (Yang, 2015; Morris and Yang, 2022; Denti, 2022), bargaining (Ravid, 2020), and attention management (Lipnowski, Mathevet, and Wei 2020). The most relevant paper is Yang (2020), who studies a security-design problem related to our model.

Several papers use more structured learning models to explore how the buyer's incentives to acquire information depends on the selling mechanism. For example, Crémer and Khalil (1992) consider a buyer who contracts with a seller who decides whether or not to pay in order to observe their cost of production, and show the seller remains uninformed at the optimum. Persico

[2]Armstrong and Zhou (2022) studies the effect of information on profits and consumer surplus in oligopolistic competition. In addition, several papers use information design tools to study information provision in markets. For example, see Hwang, Kim, and Boleslavsky (2019), Smolin (2020), and Yang (forthcoming).

(2000) shows buyers acquire less information in a second-price auction than in a first-price one, provided that their signals are affiliated. Bergemann and Välimäki (2002) shows that with information acquisition, the classic Vickrey-Clark-Groves mechanism still implements the efficient allocation when values are private, but that efficiency may fail when values are common. Compte and Jehiel (2007) show simultaneous auctions generate lower revenue than dynamic ones when buyers have an opportunity to learn. Shi (2012) characterizes the revenue-maximizing auction in private-value settings. In addition to their focus on auctions, these models differ from ours in that they require the buyer to choose among a set of signal structures that can be linearly ordered in their informativeness.[3]

2. Model

There is a monopolist (she) and a buyer (he). The game begins with the monopolist offering the buyer a menu, which is a compact set of pairs, $M \subseteq [0,\bar{q}] \times \mathbb{R}$.[4] Each menu item $(q,t) \in M$ corresponds to a transfer of t to be paid to the monopolist by the buyer, and the quality q of the product the buyer gets in exchange. The buyer's utility from (q,t) depends on his type, $\boldsymbol{\theta}$, a random variable distributed over $\Theta = [\underline{\theta},\bar{\theta}] \subseteq \mathbb{R}_+$ according to a CDF F_0. We denote the prior-expected type by $\theta_0 := \int \theta F_0(\mathrm{d}\theta)$, and assume F_0 includes $\underline{\theta}$ and $\bar{\theta}$ in its support. Given θ, the buyer's utility from (q,t) is

$$U(\theta,q,t) = \theta q - t.$$

The monopolist's payoff from the buyer's chosen menu item (q,t) is

$$\Pi(q,t) = t - \kappa(q),$$

where $\kappa : \mathbb{R}_+ \to \mathbb{R}_+$ is an increasing, continuously differentiable, and convex function satisfying $\kappa(0) = 0$. We also assume it is efficient to serve all types, meaning $\kappa'(0) < \underline{\theta}$, though this assumption can often be relaxed—see Section 6. We say **marginal quality costs are constant** if $\kappa(q) = \kappa_0 q$ for some $\kappa_0 \in \mathbb{R}_+$, and **marginal quality costs are strictly increasing** if κ is strictly convex. The monopolist's menu must give the buyer the option of not buying anything, meaning M must include the option $(0,0)$. Both the monopolist and the buyer are risk-neutral expected utility maximizers.

Neither the monopolist nor the buyer knows $\boldsymbol{\theta}$, but the buyer can choose to learn about it after observing the monopolist's menu. The buyer's information acquisition is flexible, meaning he can

[3] Another strand of the literature studies the seller's benefits from revealing information about the buyers' valuations prior to participating in an auction; see, for example, Milgrom and Weber (1982), Ganuza (2004), Bergemann and Pesendorfer (2007), Ganuza and Penalva (2010), and Li and Shi (2017).

[4] We require the menu to be compact to ensure existence of an optimal choice for the buyer.

use any signal s to learn about $\boldsymbol{\theta}$. The flexibility assumption expresses ideas. First, the buyer has access to detailed data that helps him determine his exact value for the item. Second, the buyer can collect only the information he finds useful, avoiding any unnecessary effort to gather or process data he wishes to disregard. For instance, in the context of cell phone data plans, the first idea corresponds to the buyer being able to closely monitor his data usage, understand potential scenarios affecting it, and assess his marginal value for data. The second idea says the buyer who considers a plan with a fixed monthly data allowance can calculate their average monthly usage without needing to analyze daily variations.

An alternative approach for modeling the buyer's learning decision is to restrict him to a fixed set of signals that are ordered by their informativeness. This approach is well-suited for studying the connection between the monopolist's menu and the *amount* of information the buyer decides to acquire. By contrast, models of flexible learning are appropriate for addressing questions that focus on the *kind* of information the agent wants to learn. This is the case in the current paper: our interest is in understanding how the need to steer the buyer towards learning more favorable information impacts the shape of the monopolist's menu.

The functional form of the buyer's utility means that his expected payoff from any menu item depends on his posterior mean, $\mathbb{E}\left[\boldsymbol{\theta}|\mathrm{s}\right]$. Therefore, the marginal distribution of $\mathbb{E}\left[\boldsymbol{\theta}|\mathrm{s}\right]$ pins down the buyer's expected trade surplus from any menu. This distribution also determines the probability the buyer purchases any menu item, which, in turn, is sufficient for calculating the monopolist's profits and optimal menu. In other words, trade outcomes depend only on the marginal distribution of the buyer's posterior mean, and so we identify each signal with the CDF of this marginal.[5] More precisely, letting $\mathcal{F}$ be the set of all CDFs over Θ, we let the buyer choose any element of $\mathcal{F}$ that can arise as the marginal CDF of $\mathbb{E}\left[\boldsymbol{\theta}|\mathrm{s}\right]$ for some s. We denote this set by $\mathcal{I}$ and describe it formally below.

As observed by Gentzkow and Kamenica (2016), F is the CDF of the marginal distribution of the buyer's posterior mean for some signal if and only if it is a mean-preserving contraction of the prior, F_0. Recall that $F \in \mathcal{F}$ is a **mean-preserving spread** of $G \in \mathcal{F}$ (denoted by $F \succeq G$) if and only if

$$\int_{\tilde{\theta}\leq\theta} \left(F - G\right)(\tilde{\theta})\mathrm{d}\tilde{\theta} \geq 0 \text{ for all } \theta \in \Theta, \text{ with equality at } \theta = \bar{\theta}.$$

The CDF F is a **strict mean-preserving spread** of G (denoted by $F \succ G$) if both $F \succeq G$ and $G \neq F$.[6] Therefore, one can describe the set of feasible posterior mean distributions via

$$\mathcal{I} = \left\{F \in \mathcal{F} : F_0 \succeq F\right\}.$$

[5]This method of modeling flexible information is common in the information-design literature; see, for example, Gentzkow and Kamenica (2016), Roesler and Szentes (2017), Kolotilin (2018), and Dworczak and Martini (2019).

[6]Notice $\succeq$ is reflexive and anti-symmetric, meaning $F \succeq G$ and $G \succeq F$ if and only if $F = G$.

We refer to CDFs in $\mathcal{I}$ as **signals**. Given $F \in \mathcal{I}$, we denote the lowest and highest realizations F can generate by $\underline{\theta}_F := \min[\text{supp } F]$ and $\bar{\theta}_F := \max[\text{supp } F]$, respectively.

It turns out to be convenient to describe the set $\mathcal{I}$ via a continuum of inequality constraints. For every CDF F, define the function $I_F : \Theta \to \mathbb{R}$ as

$$I_F(\theta) := \int_{\tilde{\theta} \in [\underline{\theta}, \theta]} (F_0 - F)(\tilde{\theta}) \mathrm{d}\tilde{\theta}.$$

In other words, $I_F(\theta)$ gives the difference between the integral of F_0 and the integral of F over the range $[\underline{\theta}, \theta]$. A CDF F is a signal if $I_F(\theta) \geq 0$ holds for all θ, and $I_F(\bar{\theta}) = 0$. Intuitively, $I_F(\theta)$ measures the degree to which the signal that generates F pools states above and below θ. Indeed, $I_F(\theta) = 0$ if and only if every signal structure that generates F does not pool types strictly below θ with types strictly above θ with positive probability; that is, the signal must separate types above and below θ. We follow Ravid, Roesler, and Szentes (2022) and refer to any θ with $I_F(\theta) = 0$ as F-**separating**, and refer to any θ that is not F-separating as F-**pooling**.

Information acquisition comes at a cost. In general, different information structures generating the same distribution of posterior expectations might come at different costs. However, because the buyer's expected payoff from trade depends only on the distribution of this posterior expectation, F, she would always use the least expensive signal structure that leads to F. In fact, the buyer may even randomize to get F. Thus, we can evaluate the cost of F by the expected cost of the cheapest randomization that generates it, resulting in a indirect cost function, $C : \mathcal{I} \to \mathbb{R}_+$. We follow Ravid, Roesler, and Szentes (2020) and state our assumptions directly in terms of this C. We assume C is continuous, affine,[7] and strictly increasing in informativeness; that is, $C(F) > C(F')$ whenever F is a strict mean-preserving spread of F'. In the online appendix, we prove these properties imply the existence of some continuous, strictly convex function $c : \Theta \to \mathbb{R}_+$ such that

$$C(F) = \int c(\theta) F(\mathrm{d}\theta).$$

Moreover, we show it is without loss for c to attain its minimum at θ_0. In addition, we require c to be a twice differentiable function with a strictly positive second derivative.

After choosing F, the buyer gets to see its realization, $\theta \in \Theta$, and decides whether to purchase, and if so, what item to select from the menu to maximize his expected utility.

To summarize, the game begins with the monopolist choosing a menu. Next, the buyer observes the menu, and chooses what signal $F \in \mathcal{I}$ to acquire. The buyer then sees his signal realization

[7] Some readers may be interested in deriving these properties from a more primitive object, such as cost function defined over the distribution of the agent's posterior belief. To do so, one can assume the cost of a distribution over posteriors is posterior separable (Caplin, Dean, and Leahy, 2021), with a divergence function that depends only on posterior's mean. Mensch and Malik (2023) provides a revealed preference characterization of such cost functions.

$\theta \in \Theta$, and chooses an item from the monopolist's menu. We are interested in the menu that maximizes the monopolist's expected profits, subject to the buyer behaving optimally, which exists by the following theorem.

Theorem 1. *A monopolist-optimal menu exists.*

Our timing assumptions mean the buyer's interim expected payoff is fully determined by her posterior mean. Hence, this mean completely determines the buyer's decision from the monopolist's menu. As such, by the revelation principle, it is sufficient to focus on direct revelation mechanisms in which the buyer reports his posterior mean. Such menus can be described with two maps,

$$Q : \Theta \to [0, \bar{q}] \, , \; T : \Theta \to \mathbb{R}_+,$$

where $Q(\theta)$ and $T(\theta)$ correspond to the quality and transfer pair chosen by a buyer with posterior mean θ. These mappings must satisfy the standard incentive compatibility and individual rationality constraints,[8]

$$\theta Q(\theta) - T(\theta) \geq \theta Q(\theta') - T(\theta') \quad \forall \theta, \theta' \in \Theta, \tag{IC}$$

$$\theta Q(\theta) - T(\theta) \geq 0 \quad \forall \theta \in \Theta. \tag{IR}$$

Usual envelope-style reasoning (Myerson, 1981) delivers that a Q and T satisfy the above two conditions if and only if Q is increasing and

$$T(\theta) = \theta Q(\theta) - \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta}) \mathrm{d}\tilde{\theta} - \underline{u}, \tag{1}$$

where $\underline{u} \geq 0$ is the utility granted to the lowest possible type,

$$\underline{u} = \underline{\theta} Q(\underline{\theta}) - T(\underline{\theta}) \, .$$

It follows that $\underline{u}$ and Q are sufficient for pinning down every feasible IC and IR menu. Let $\mathcal{Q}$ be the set of all increasing functions from Θ to $[0, \bar{q}]$. We refer to a $Q \in \mathcal{Q}$ as an **allocation**, and (with slight abuse of terminology) we refer to $(Q, \underline{u}) \in \mathcal{Q} \times \mathbb{R}_+$ as a **mechanism**. Given a mechanism $(Q, \underline{u})$, we let $T_{Q,\underline{u}}$ denote the transfer implied by (1). This description implies that a type-θ buyer's utility from truthful reporting under $(Q, \underline{u})$ is

$$V_{Q,\underline{u}}(\theta) := \underline{u} + \theta Q(\theta) - T_{Q,\underline{u}}(\theta) = \underline{u} + \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta}) \mathrm{d}\tilde{\theta}.$$

[8] Note that to accommodate deviations in the buyer's choice of signal, we must impose the constraints (IC) and (IR) even for posterior means that do not arise under the buyer's chosen CDF.

The buyer's **net value** is equal to her utility from truthful reporting θ minus the cost,

$$N_{Q,\underline{u}} := V_{Q,\underline{u}}(\theta) - c(\theta).$$

Given an allocation Q, let $\underline{\theta}_Q$ be the highest θ type that Q excludes, and $\bar{\theta}_Q$ to be the lowest type to which Q gives the highest quality. If no type is excluded (gets the highest quality), set $\underline{\theta}_Q = \underline{\theta}$ ($\bar{\theta}_Q = \bar{\theta}$).[9]

We now state the monopolist's problem of choosing a profit-maximizing mechanism. Given a mechanism $(Q, \underline{u})$, the buyer's utility from using signal $F \in \mathcal{I}$ is given by his expected net value, $\int N_{Q,\underline{u}}(\theta) F(\mathrm{d}\theta)$. We refer to a mechanism-signal tuple $(Q, \underline{u}, F)$ as **an outcome**, and say the outcome is **incentive compatible** (IC) if F maximizes the buyer's utility given $(Q, \underline{u})$,

$$F \in \underset{\tilde{F} \in \mathcal{I}}{\operatorname{argmax}} \int N_{Q,\underline{u}}(\theta) \tilde{F}(\mathrm{d}\theta). \tag{2}$$

Consistent with this terminology, whenever $(Q, \underline{u}, F)$ is IC, we say $(Q, \underline{u})$ is F**-incentive compatible** (F-IC). Denote the monopolist's payoff when the buyer reports a signal realization of θ by

$$\pi_{Q,\underline{u}}(\theta) := T_{Q,\underline{u}}(\theta) - \kappa(Q(\theta)) = \theta Q(\theta) - V_{Q,\underline{u}}(\theta) - \kappa(Q(\theta)).$$

Then, we can write the monopolist's expected profit from using offering $(Q, \underline{u})$ when the buyer uses F as $\int \pi_{Q,\underline{u}}(\theta) F(\mathrm{d}\theta)$, and so the monopolist's program is given by

$$\max_{(Q,\underline{u},F)} \int \pi_{Q,\underline{u}}(\theta) F(\mathrm{d}\theta) \text{ s.t. } (Q, \underline{u}, F) \text{ is IC.}$$

The goal of this paper is to study the above program.

Before proceeding, we make an observation that simplifies the analysis. To state this observation, we first introduce some definitions. Fix an increasing function $\varphi : \Theta \to [x, y]$. We say φ is **constant around** θ whenever it is constant in some open neighborhood of θ. If φ is not constant around θ, we say that φ is **strictly increasing** at θ. For every $\theta \in (\underline{\theta}, \bar{\theta}]$, let $\varphi_-(\theta) = \sup_{\theta' < \theta} \varphi(\theta')$ be the left limit of φ at θ, and set $\varphi_-(\underline{\theta}) = x$. Similarly, for $\theta \in [\underline{\theta}, \bar{\theta})$, define the right limit of φ and θ by $\varphi_+(\theta) = \inf_{\theta' > \theta} \varphi(\theta')$, while setting $\varphi_+(\bar{\theta}) = y$. Relatedly, for a convex function $\phi : \Theta \to [x, y]$, we let we let ϕ'_- and ϕ'_+ denote its left and right derivatives, respectively, whenever those exist (which is the case for every $\theta \in (\underline{\theta}, \bar{\theta})$).

Armed with the above definitions, we say that an allocation Q **jumps towards efficiency** if

$$Q(\theta) \in \underset{q \in [Q_-(\theta), Q_+(\theta)]}{\operatorname{argmax}} [\theta q - \kappa(q)]. \tag{3}$$

[9]Formally, set $\bar{\theta}_Q := \sup \left[\{\underline{\theta}\} \cup Q^{-1}[0, \bar{q})\right]$ and $\underline{\theta}_Q = \inf \left[\{\bar{\theta}\} \cup Q^{-1}(0, \bar{q}]\right]$.

In the appendix we prove two things. First, we show one can take any allocation and replace it with an allocation that jumps towards efficiency without reducing the monopolist's profits. Consequently, focusing on allocations that jump towards efficiency is without loss of optimality. Second, we show that allocations that jump towards efficiency convey a technical benefit: if Q jumps towards efficiency, then $\pi_{Q,\underline{u}}$ is an upper-semicontinuous function. Therefore, we restrict attention to allocations that jump towards efficiency for the rest of the paper.

3. Cost-Canceling Allocations

In this section, we build on Dworczak and Martini's (2019) duality results to obtain necessary and sufficient conditions for some F to be optimal for the buyer. Using these conditions, we show it is without loss to restrict the monopolist to a convenient class of allocations, which we call cost-canceling. These allocations enable us to relate the monopolist optimal outcome to two familiar maximization programs: one program that involves maximization over allocations, and the other program that optimizes over the buyer's information.

We begin with characterizing the buyer's optimal signal. The key to our characterization is the concept of an F**-marginal price**, which is an increasing function

$$p : \Theta \to [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$$

that is constant around any θ that is not F-separating. The following result shows that marginal price functions are tightly connected to the solution of the buyer's information acquisition problem.

Lemma 1. *Fix a mechanism $(Q, \underline{u})$. Then F solves (2) if and only if an F-marginal price p exists such that the function*

$$P_{Q,\underline{u},p}(\theta) := N_{Q,\underline{u}}(\underline{\theta}_F) + \int_{\underline{\theta}_F}^{\theta} p(\tilde{\theta})\mathrm{d}\tilde{\theta} \tag{4}$$

lies weakly above $N_{Q,\underline{u}}(\theta)$ for all θ, and equals $N_{Q,\underline{u}}(\theta)$ for all θ in the support of F. Moreover, one can choose p such that $p(\theta) = Q(\theta) - c'(\theta)$ for all $\theta \in \operatorname{supp} F$.

The proof of the above lemma relies heavily on Dworczak and Martini's (2019) duality-based approach for solving mean-measurable information design problems. In the context of our model, this result can be stated as follows:[10] given the mechanism $(Q, \underline{u})$, the signal F solves the buyer's problem (2) if and only if a Lipschitz continuous and convex function $P : \Theta \to \mathbb{R}$ exists that satisfies the following two properties:

1. every θ has $P(\theta) \geq N_{Q,\underline{u}}(\theta)$, with equality holding F-almost surely, and

[10] To apply the result to the case where c' is unbounded, one needs to use Dizdar and Kováč's (2020) generalization of Dworczak and Martini (2019). See the Appendix for precise formulation.

2. the function P is affine over any interval (θ_1, θ_2) of F-pooling types.[11]

Dworczak and Martini (2019) refer to the function P as a *price function*. Intuitively, one can think of P as being the outcome of a competitive equilibrium in an economy with a representative consumer whose utility is given by $N_{Q,\underline{u}}$, and a production technology that generates outputs using mean-preserving spreads. Consequently, $P(\theta)$ gives the value of the optimal way of creating mean preserving spreads and mean-preserving contractions that involve θ.

The key observation behind Lemma 1 is that marginal price functions are just (almost everywhere) derivatives of price functions. Indeed, the requirement that marginal price functions are increasing corresponds to convexity of price functions, whereas the restriction that marginal prices are constant around signal realizations that are F-pooling corresponds to price functions being affine over the same region. Consequently, one can use $P_{Q,\underline{u},p}$ as a price function certifying the optimality of F whenever $P_{Q,\underline{u},p}$ satisfies the lemma's desiderata. Conversely, whenever F is optimal, one can obtain an F- marginal price function satisfying the lemma's requirements by taking a derivative of the price function delivered by Dworczak and Martini (2019).[12]

We are now ready to define the notion of information-cost-cancelling allocations. Fix some signal F and some F-marginal price p. Define the allocation Q^p via [13]

$$Q_p(\theta) = \min\left\{(p(\theta) + c'(\theta))_+, \bar{q}\right\} = \begin{cases} p(\theta) + c'(\theta), & \theta \in (\underline{\theta}_F, \bar{\theta}_F) \\ \max\{p(\underline{\theta}_F) + c'(\theta), 0\}, & \theta \in [\underline{\theta}, \underline{\theta}_F] \\ \min\{p(\bar{\theta}_F) + c'(\theta), \bar{q}\}, & \theta \in [\bar{\theta}_F, \bar{\theta}] \end{cases} \tag{5}$$

We say an allocation Q is F**-information-cost-canceling** (F-ICC) if $Q = Q_p$ for some F-marginal price p. If Q is F-ICC, we let p_Q be the F-marginal price for which $Q = Q_{p_Q}$. We refer to a mechanism as F**-information-cost-canceling** if its allocation is $F-$ICC. Finally, we say an allocation is information cost canceling (ICC) if it F-ICC for some F.

In Figure 1, we illustrate the construction of an F-ICC allocation for the case in which $\Theta = [0, 1]$, the learning cost c is given by entropy,

$$c(\theta) = \theta \ln(\theta) + (1 - \theta) \ln(1 - \theta),$$

and θ^* is the only F-separating type; that is, θ^* is the only F-separating type in $(0, 1)$. The left panel depicts c' and an F-marginal price p. Observe p is constant on the intervals $(0, \theta^*)$ and

[11] Requiring P to be affine over intervals of F-pooling types is equivalent to Dworczak and Martini's (2019) requirement that $\int P(\theta)(F - F_0)(\mathrm{d}\theta) = 0$.

[12] Our proof actually relies on Dizdar and Kováč's (2020) generalization of the Dworczak and Martini's (2019) result—see the Appendix for exact details.

[13] For $x \in \mathbb{R}$, we use the convention $(x)_+ = \max\{x, 0\}$.

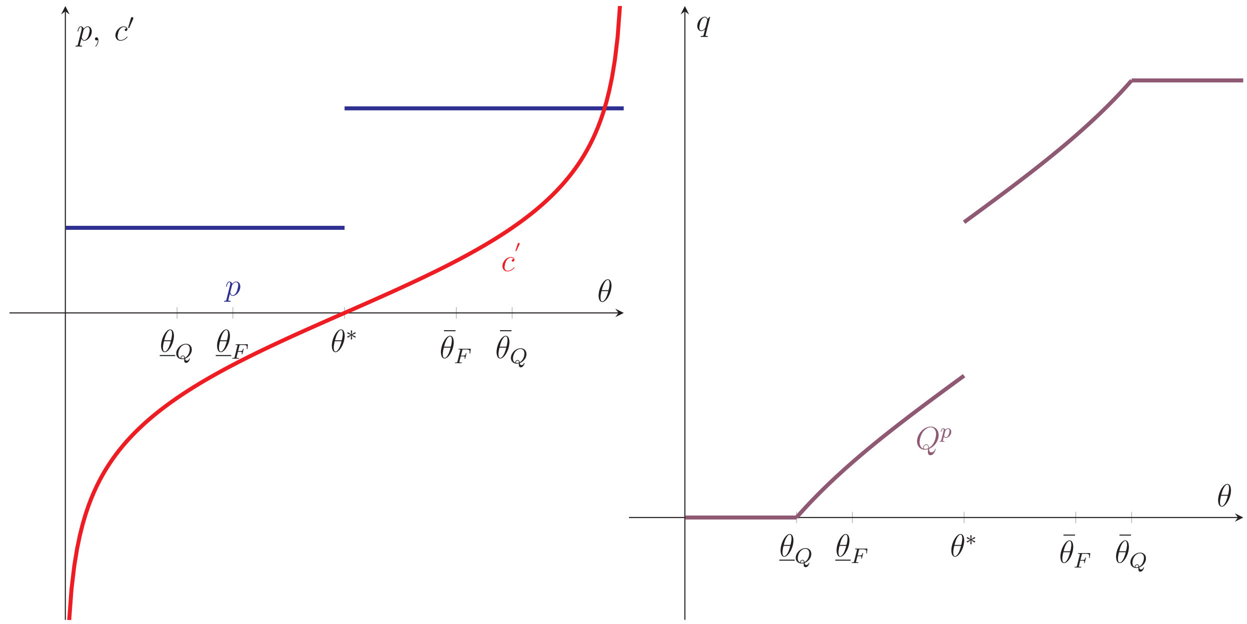

Figure 1: Construction of an F-ICC allocation for an F that satisfies $I_F(\theta^*) = 0$.

$(\theta^*, 1)$, where I_F is strictly positive. By the definition of an F-ICC allocation, Q_p is given by (5), as illustrated in the right panel. Note that Q_p is constant for sufficiently low θ, where $p(\theta)+c'(\theta)$ is negative; similarly, Q_p stays constant once $p(\theta)+c'(\theta)$ hits $\bar{q}$. In between, Q_p is strictly increasing since c' is strictly increasing and p is weakly increasing. Finally, at θ^*, both p and Q_p jump by the same amount.

The motivation behind the ICC definition is as follows. Suppose the mechanism $(Q, \underline{u})$ is F-IC—i.e., F solves the buyer's problem. By Lemma 1, we can find a marginal price p with the property that $P_{Q,\underline{u},p}(\theta) \geq N_{Q,\underline{u}}(\theta)$ holds for all θ, with equality holding on the support of F. If, however, $Q = Q_p$, the equality $P_{Q,\underline{u},p}(\theta) = N_{Q,\underline{u}}(\theta)$ holds not only on the support of F: it holds for all θ in $[\underline{\theta}_Q, \bar{\theta}_Q]$. Figure 2 illustrates this property for the F-marginal price from Figure 1. Figure 2 shows the price function $P_{Q_p,0,p}$ as well as the agent's net value from a given signal realization θ, $N_{Q_p,0}$.

Economically, ICC allocations require the change in the accumulation rate of the buyer's information rents over $[\underline{\theta}_F, \bar{\theta}_F]$ to satisfy two bounds: the change must be at least as high as the change in his marginal learning costs, and strictly higher only over F-separating regions. These two bounds are constructed in order to satisfy the buyer's incentive to acquire F. The lower bound guarantees N is is not concave, meaning that the buyer does not benefit from conducting mean-preserving contractions. The upper-bound dissuades the buyer from conducting mean-preserving spreads by ensuring that N is only strictly convex over intervals where such spreads are infeasible.

Next, we show that focusing on F-ICC mechanisms is without loss. Moreover, every F-ICC

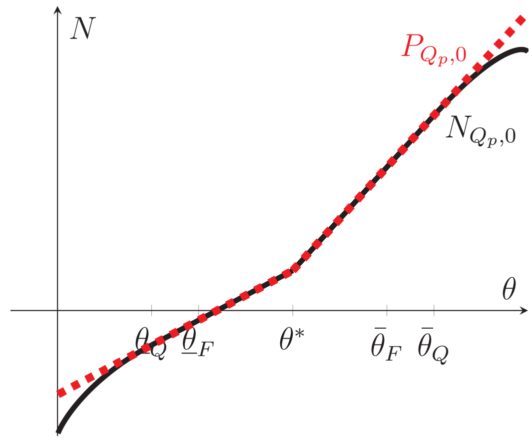

Figure 2: Value from F-ICC allocation for F-price P

is F-IC—that is, an F-ICC mechanism always makes F optimal for the buyer.

Theorem 2. *Every F-ICC mechanism is F-IC. Moreover, if $\left(\tilde{Q}, \tilde{\underline{u}}\right)$ is an F-IC mechanism, then an F-ICC mechanism $(Q, \underline{u})$ exists such that $\underline{u} \geq \tilde{\underline{u}}$, and both $Q(\theta) = \tilde{Q}(\theta)$ and $V_{Q,\underline{u}}(\theta) = V_{\tilde{Q},\tilde{\underline{u}}}(\theta)$ hold for all $\theta \in \text{supp}F$.*

That every F-ICC mechanism $(Q, \underline{u})$ is F-IC follows from Lemma 1 and the above-mentioned property of F-ICC mechanisms, namely that $N_{Q,\underline{u}}$ lies below $P_{Q,\underline{u},p_Q}$, with the two being equal on $[\underline{\theta}_Q, \bar{\theta}_Q]$. To get that every F-IC mechanism can be replaced with an equivalent F-ICC mechanism, we use Lemma 1 to obtain an F-marginal price function p that certifies the optimality of F, and show that $Q_p = Q$ over the support of F. We then define $\tilde{\underline{u}}$ to guarantee that $V_{Q,\underline{u}}(\theta) = V_{Q_p,\tilde{\underline{u}}}(\theta)$ holds for all $\theta \in \text{supp}F$.

We now explain that the restriction to information-cost cancelling allocations is useful because it enables us to reason about the monopolist's problem "one dimension at a time". Specifically, we note the monopolist optimal allocation must solve a familiar-looking one-dimensional mechanism design program, whereas the optimal signal must solve a particular mean-measurable information design problem. Thus, one can use familiar techniques to make inferences about the monopolist's optimal outcome. Towards a formal statement of this observation, note first that setting $\underline{u} = 0$ is always optimal for the monopolist. Therefore, hereafter we abuse notation, writing $\pi_Q := \pi_{(Q,0)}$, $V_Q := V_{Q,0}$, and $N_Q := N_{Q,0}$, and use (Q, F) to refer to the outcome $(Q, 0, F)$.

Corollary 1 below outlines the above-mentioned programs. The starting point for both programs is a monopolist optimal outcome (Q, F). The mechanism design problem is based on the observation that, holding F fixed, the allocation Q must maximize the monopolist's profit across all F-IC allocations. However, since each such allocation is outcome equivalent (under F) to some F-ICC allocation, maximizing profit across all F-IC allocations is equivalent to maximizing profit across all F-ICC allocations. The mechanism design program then follows from observing that

F-ICC allocations are indexed by their corresponding F-marginal price functions. Denote the set of all F-marginal price functions by $\mathcal{P}(F)$.

A similar logic delivers the information design program: holding Q fixed, the signal F must maximize the monopolist's profit across all signals that are IC for the buyer given Q. In particular, if Q is F-ICC, the signal F must be better for the monopolist than any other signal for which Q is ICC. Given an ICC allocation Q, we let $\mathcal{I}(Q)$ be the set of signals for which Q is ICC.[14] It is straightforward to verify that $\mathcal{I}(Q)$ is a compact and convex set.

Corollary 1. *Suppose (Q, F) is monopolist optimal. For any F-ICC allocation $\tilde{Q}$, the outcome $(\tilde{Q}, F)$ is monopolist optimal if and only if $p_{\tilde{Q}}$ solves*

$$\max_{p \in \mathcal{P}(F)} \int \pi_{Q_p}(\theta) F(\mathrm{d}\theta). \tag{6}$$

Moreover, if Q is F-ICC, then $(Q, \tilde{F})$ is monopolist optimal if and only if

$$\max_{\tilde{F} \in \mathcal{I}(Q)} \int \pi_Q(\theta) \tilde{F}(\mathrm{d}\theta). \tag{7}$$

Thus, once we know the monopolist optimal signal, finding the monopolist optimal allocation amounts to finding the F-marginal price that solves the program (6). Similarly, once we know the monopolist optimal ICC allocation, finding the optimal signal amounts to maximizing the monopolist's profit across all signals for which that allocation is ICC. Next, we use these two programs to make inferences about the monopolist optimal outcome.

4. Downward Quality Distortions

In this section we use the reduction to F-ICC allocations to analyze the efficiency of the monopolist optimal allocation. We prove two main results. First, we show that when marginal costs of quality are strictly increasing, the monopolist optimal allocation distorts quality downward for all of the buyer's realized type. Notably, this downward distortion is strict for all realizations below $\bar{\theta}$, the highest possible under the prior. Thus, whenever the optimal outcome involves a signal whose highest realization lies strictly below $\bar{\theta}$, one gets that quality is strictly distorted downwards for all posterior means that can realize under the buyer's chosen signal structure. In a sense, this result reverses the famous "no distortion at the top" observation from the exogenous information case. Second, we show that with constant marginal costs of quality, the highest realized θ always gets the efficient quality of $\bar{q}$. All other types getting a strictly lower quality. Again, this result reverses

[14]That is, $\mathcal{I}(Q)$ is the set of all signals $F \in \mathcal{I}$ such that $\mathrm{supp}(F) \subseteq [\underline{\theta}_Q, \bar{\theta}_Q]$, and $I_F(\theta) > 0$ only if p_Q is constant at θ.

a well-known result from the exogenous information case, whereby the monopolistic optimal allocation involves serving the efficient quality to all types above some (typically interior) threshold, while excluding all other types.

We begin with Theorem 3, which considers the case with increasing marginal costs for quality.

Theorem 3. *Suppose κ is strictly convex. Every monopolist optimal outcome (Q^*, F^*) admits an allocation Q such that $Q = Q^*$ holds F^*-almost surely, (Q, F^*) is monopolist optimal, and every $\theta \in \operatorname{supp} F^*$ has $\theta \geq \kappa'(Q(\theta))$, with equality holding if and only if $\theta = \bar{\theta}$.*

The above theorem has an immediate corollary: whenever learning costs are sufficiently steep at the top, the monopolist finds it optimal to provide a quality that is below efficient even to the highest realized type.

Corollary 2. *Suppose κ is strictly convex, and $c'(\bar{\theta}) - c'(\theta_0) > \bar{q}$. Every monopolist optimal outcome (Q^*, F^*) admits an allocation Q such that $Q = Q^*$ holds F^*-almost surely, (Q, F^*) is monopolist optimal, and $\theta > \kappa'(Q(\theta))$ for all $\theta \in \operatorname{supp} F^*$.*

The reasoning behind the corollary is straightforward. Whenever $c'(\bar{\theta}) - c'(\theta_0) > \bar{q}$, every ICC allocation Q must have $\bar{\theta}_Q < \bar{\theta}$. Theorem 2 then implies that a signal F can be made incentive compatible with some allocation only if $\bar{\theta}_F < \bar{\theta}$. The result then follows from Theorem 3.

Together, Theorem 3 and Corollary 2 highlight a fundamental difference between the exogenous and endogenous information cases. When information is exogenous, Mussa and Rosen's (1978) analysis shows that quality is distorted downward at all types *except* the highest one. Our results demonstrate that, when information is endogenous, quality is still distorted downwards, but the "no distortion" at the top remains if and only if the highest type under the buyer's chosen signal is the highest possible type when the buyer is fully informed. Consequently, we obtain that the "no distortion at the top" result fails whenever the buyer chooses to never learn that her type equals $\bar{\theta}$, a situation that occurs whenever learning costs are sufficiently steep.

The theorem's logic relies on two economic forces. The first force is familiar from Mussa and Rosen's (1978) model of monopolistic screening with exogenous information. In that model, the monopolist distorts the allocation of lower types downwards in order to reduce the information rents given to higher types. Using the optimal allocation program (6) from Corollary 1, we show a similar concern arises in the current setting for any type at which the monopolist's chosen marginal price is strictly increasing.

The second force comes from the endogenous nature of the buyer's information. By having the buyer conduct incentive-compatible mean-preserving spreads and contractions, the monopolist can trade-off the benefit of having different mean-realizations. In particular, it turns out that whether or not the monopolist benefits from marginal increase the buyer's realized posterior mean depends on whether or not that mean's quality is above or below the efficient level. Roughly speaking, this relationships follows from noting that an increase in the buyer's mean-realization has two effects on the monopolist's profits: First, it changes the total surplus generated by the transaction, and second, it impacts the information rents the monopolist cedes to the buyer. By the envelope theorem, the change in the buyer's information rents is second order, meaning the difference in the available social surplus dominates. Since the provided quality is increasing with the buyer's type, a small increase in θ raises social surplus if and only if quality is under-provided at θ. Thus, π_Q increases at θ if and only if $Q(\theta)$ is below the efficient level.[15]

To see the relevance of the observation to Theorem 3, take any $\theta \in \operatorname{supp} F$ with $\theta < \bar{\theta}$ around which p_Q is constant. Let $\tilde{\theta}$ be the highest type below θ at which p_Q is strictly increasing. Suppose, for the sake of intuition, that both θ and $\tilde{\theta}$ occur with positive probability under F. Consider the change in profits due to a small mean-preserving contraction that replaces each realization of θ with a slightly lower realization, and each realization of θ' with a slightly higher one. By choice of $\tilde{\theta}$, p_Q must be constant over $[\tilde{\theta}, \theta]$, meaning this contraction is incentive compatible for the buyer. Therefore, because F is optimal, the total effect of this contraction must be negative. Recall, however, that because p is increasing at $\tilde{\theta}$, the quality $Q(\tilde{\theta})$ must below its efficient level. Hence, increasing $\tilde{\theta}$ slightly must strictly increase the seller's profit, as explained above. Since this contraction cannot be profitable, we get that the slight reduction in θ must strictly decrease the monopolist's profits, and so θ's quality must be strictly below its efficient level.

Next, we discuss the quality inefficiencies introduced when marginal quality costs are constant—i.e., when $\kappa(q) = \kappa_0 q$. In the full-learning case, the monopolist optimal allocation involves providing the quality of $\bar{q}$ to all θ about some (typically interior) threshold, with all other θ's being excluded (e.g., Riley and Zeckhauser, 1983). Consequently, the monopolist provides the efficient quality to all types above the threshold, and a severe under provision of quality otherwise.

Our next result shows that taking the buyer's learning into account results in only the top θ getting the efficient (i.e., maximal) quality. Moreover, only the lowest possible posterior mean according to the buyer's chosen information structure is ever excluded. All other posterior means must receive an interior quality.

Corollary 3. *Suppose marginal quality costs are constant. If (Q, F) is monopolist optimal, then $Q(\theta) = \bar{q}$ if and only if $\theta \geq \bar{\theta}_F$. Moreover, we have $Q(\theta) > 0$ for every $\theta > \underline{\theta}_F$.*

Thus, among interior θ's, the need to provide the buyer with incentives to purchase her information reduces quality provided to the higher posterior means, but increases the quality given to

[15]More formally, suppose Q is an F-ICC allocation such that p_Q is constant on $[\theta_1, \theta_2]$. One can show that, in this case, the right derivative $\pi'_{Q+}(\theta_1)$ and the left derivative $\pi'_{Q-}(\theta_2)$ are well defined, and that $\pi'_{Q+}(\theta)$ $(\pi'_{Q-}(\theta))$ is strictly positive (negative) if and only if $Q(\theta)$ is strictly below (above) the efficient level.

the lower means. This distortion stems from the need to provide the buyer with incentives to purchase her information. Intuitively, the buyer will only acquire a signal that generates two different realizations θ and θ' if they result in different choices from the monopolist's menu. Consequently, a posterior mean θ that lies strictly between $\underline{\theta}_F$ and $\bar{\theta}_F$ must receive an interior quality. Hence, to provide the buyer with incentives to acquire the information that leads to different realizations, the monopolist must reduce the quality provided to interior types below the efficient level.

5. Bi-Poolings, Steep Learning-Costs, and Simple Menus

Next, we make inferences about the buyer's information in the monopolist optimal outcome using the first program outlined in Corollary 1. This program is similar to the standard mean-measurable persuasion problem studied by Kolotilin (2018), Dworczak and Martini (2019) and others. The main difference between those programs and the program we outline in (7) is that, in (7), the buyer's information must be incentive compatible—i.e., F must be in $\mathcal{I}(Q)$. Despite this constraint, many of the techniques developed for solving the standard mean-measurable problem are also valid for for solving (7). Consequently, one can use these techniques learn about the buyer's signal at the optimum. In particular, we show it is always optimal to have the buyer acquire a "bipooling" signal (Arieli et al., 2023). We then take this fact and combine it with the restriction to ICC allocations to get that a menu with a single purchasing option is optimal whenever learning costs are steep, and one of two conditions hold: either the monopolist always wants the buyer to learn as much as possible, or the monopolist always prefers that the buyer obtains no information.

Throughout this section, we assume F_0 is a continuous distribution with full support. This assumption can be relaxed easily by generalizing Arieli et al.'s (2023) bipooling result to priors with atoms and gaps in their support. Such a generalization is straightforward, but tedious, and so we leave it out of our paper for the sake of brevity.

Let us proceed with the analysis. Examining Corollary 1 immediately reveals the following implication: to obtain a monopolist optimal outcome, one can pair an optimal ICC allocation Q with *any* solution to program (7). Note, however, that this program involves maximizing a linear objective over the compact convex set $\mathcal{I}(Q)$. We can therefore appeal to Bauer's Maximum Principle and obtain that the monopolist optimal F must be an extreme point of $\mathcal{I}(Q)$. Hence, one can learn more about the shape of the buyer's signal by exploring the extreme points of the set $\mathcal{I}(Q)$.

Below we show $\mathcal{I}(Q)$'s extreme points must be a subset of the bipooling family (Arieli et al., 2023). Formally, a distribution $F \in \mathcal{I}$ is a **bi-pooling** if a (potentially empty) collection of disjoint open intervals $((\underline{\theta}_n, \bar{\theta}_n))_{n=1}^{N}$ for $N \in \{0, 1, \ldots\} \cup \{\infty\}$ exists such that

(i) every $\theta \notin \cup_n (\underline{\theta}_n, \bar{\theta}_n)$ is F-separating,

(ii) for every n, $F_-(\bar{\theta}_n) - F(\underline{\theta}_n) = F_{0-}(\bar{\theta}_n) - F_0(\underline{\theta}_n)$ and $\left|\text{supp}(F) \cap (\underline{\theta}_n, \bar{\theta}_n)\right| \leq 2$.

In other words, a bi-pooling is the posterior mean distribution generated by a signal such that every valuation θ outside $\cup_n(\underline{\theta}_n, \bar{\theta}_n)$ is revealed. For other θ, the signal reveals which interval θ belongs to, as well as an additional binary signal. Arieli et al. (2023) show the set of bi-poolings coincides with the set of extreme points of $\mathcal{I}$.

The following proposition shows the monopolist finds it optimal to have the buyer acquire a bi-pooling signal.

Proposition 1. *A monopolist optimal outcome exists in which F is a bi-pooling.*

To prove the proposition, we show that $\mathcal{I}(Q)$ is a face of the set $\mathcal{I}$; that is, we show no element of $\mathcal{I}(Q)$ can be written as a convex combination of two elements of $\mathcal{I}$ that lie outside of $\mathcal{I}(Q)$. Consequently, every extreme point of $\mathcal{I}(Q)$ must be an extreme point of $\mathcal{I}$. The result then follows from Bauer's maximum principle and Corollary 1. We note this argument goes through even if F_0 violates the stated conditions, but with bipoolings being replaced by the set of extreme points of $\mathcal{I}$. Thus, when F_0 is binary, one can replace the bipooling requirement with the requirement that the signal has binary support.[16]

One special class of bipooling signals are ones with a binary support. Our next result provides conditions under which such signals are without loss of optimality. The first condition is that learning costs are sufficiently steep. To state this condition, given some $\theta^c \in (\underline{\theta}, \bar{\theta})$, we say that F is a θ^c-**cutoff signal** if F is the posterior mean distribution induced by learning whether or not θ is strictly above θ^c.[17] We say learning costs satisfy the **steep-slope condition** if every cutoff signal F has $c'(\bar{\theta}_F) - c'(\underline{\theta}_F) > \bar{q}$. As Proposition 2 notes, the steep-slope condition leads to the emergence of binary signals at the optimum.

The second set of conditions that lead to optimality of binary signals relies on concavity of the monopolist's realized profit function. To state this condition precisely, for every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, define the function $\hat{\pi}_{\underline{\theta}^*} : [\underline{\theta}^*, \bar{\theta}] \to \mathbb{R}$ via

$$\hat{\pi}_{\underline{\theta}^*}(\theta) = \theta(c'(\theta) - c'(\underline{\theta}^*)) - \kappa(c'(\theta) - c'(\underline{\theta}^*)) - [c(\theta) - c(\underline{\theta}^*)] \,.$$

The motivation for this definition is as follows: whenever Q is an ICC allocation whose induced marginal price p_Q is constant, then π_Q equals to $\hat{\pi}_{\underline{\theta}_Q} + c(\underline{\theta}_Q)$ on $[\underline{\theta}_Q, \bar{\theta}_Q]$. Consequently, one can learn about the monopolist optimal signal by looking at the curvature of $\hat{\pi}$. Indeed, the next proposition says that concavity of $\hat{\pi}$ and marginal costs of quality being constant are together sufficient for the optimality of a binary signal.

[16]The generalization of bipoolings to a continuous, but not full support F_0 is given in Kleiner, Moldovanu, and Strack (2021).

[17]Equivalently, the support of F has one or two elements, and $I_F(\theta^c) = 0$.

Proposition 2. *A monopolist optimal outcome exists in which the buyer's signal has at most two realizations whenever one of the following conditions hold:*

(i) Learning costs satisfy the steep-slope condition.

(ii) The function $\hat{\pi}_{\underline{\theta}}$ is concave, and marginal quality costs are constant.

Let us sketch the proposition's proof. We begin with sufficiency of condition (i), which follows from a few observations. First, a signal F can be made IC under some Q if and only if the allocation Q is equal F-almost surely to some F-ICC allocation (Theorem 2). Second, every F-ICC allocation Q must satisfy $Q(\bar{\theta}_F) - Q(\underline{\theta}_F) \geq c'(\bar{\theta}_F) - c'(\underline{\theta}_F)$. This inequality is apparent from our definition of F-ICC mechanisms. Since Q takes values in $[0, \bar{q}]$, we must have $\bar{q} \geq Q(\bar{\theta}_F) - Q(\underline{\theta}_F)$. It follows that an F for which $c'(\bar{\theta}_F) - c'(\underline{\theta}_F) > \bar{q}$ is not IC under any allocation, and so cannot arise in monopolist optimal outcome. Third, mean preserving spreads make the the most extreme signal realizations more extreme; that is, $F \succeq G$ only if $\underline{\theta}_F \leq \underline{\theta}_G \leq \bar{\theta}_G \leq \bar{\theta}_F$.[18] Because c' is increasing, we obtain that when learning costs satisfy the steep-slope condition, mean preserving spreads of cutoff signals are never IC, and so cannot arise at the optimum. The fourth and final observation is that every bipooling signal with more than two signal realizations is a mean preserving spread of a cutoff signal.[19] It follows that a bipooling can arise at the optimum only if it has two or less realizations.

To prove that Proposition 2-(ii) is sufficient for optimality of a binary signal, we first prove Lemma 2 below. This lemma says that when marginal quality costs are constant, the monopolist optimal F-marginal price p is constant except for a single jump at some θ^*. At this θ^*, the marginal price p jumps from its lowest value of $p(\underline{\theta}_{Q_p})$ to $\bar{q} - c'(\bar{\theta}_F)$, which is the highest possible value any F-marginal price can take.

Lemma 2. *Suppose κ is affine, and fix some $F \in \mathcal{I}$. The program (6) admits a solution p^* that is strictly increasing at one point at most, and $p^*(\Theta) \subseteq \{p(\underline{\theta}_{Q_p}), \bar{q} - c'(\bar{\theta}_F)\}$.*[20]

The proof of Lemma 2 relies on similar logic to the extreme-point based argument of the classic "posted-price" result (see for example Manelli and Vincent, 2007). This argument is based on noting that V_Q is an affine function of the allocation Q, and so the monopolist's objective is also affine in Q. Thus, the monopolist's objective is an affine of Q, meaning Bauer's maximum theorem applies: the monopolist's objective admits an extreme point of the set of incentive compatible

[18]See Lemma 9 in Ravid, Roesler, and Szentes (2022).

[19]To obtain the fourth observation, note that a bipooling signal has more than two realizations only if it has an interior separating point. The observation then follows from noticing that θ^c is F-separating if and only if F is a mean preserving spread of the θ^c-cutoff signal.

[20]We note this lemma does not rely at all on this section's maintained assumption that F_0 is a continuous distribution with full-support.

allocations as a maximizer. The "posted-price" result then follows from noting all extreme points of this set have a "single-step" form.

To apply a similar extreme point argument to prove Lemma 2, a complication arises: whereas V_{Q_p} is affine in the allocation Q_p, it is not an affine function of the F-marginal price p that induces Q_p. Consequently, one cannot directly apply Bauer's theorem. To circumvent this issue, we identify a compact convex subset $\tilde{\mathcal{P}}$ of F-marginal prices that satisfies three properties. First, the set $\tilde{\mathcal{P}}$ includes a solution to the monopolist's problem. Second, the monopolist's objective is affine over $\tilde{\mathcal{P}}$. And third, all extreme points of $\tilde{\mathcal{P}}$ take the form of a single-step. The result then follows from applying Bauer's maximum principle to $\tilde{\mathcal{P}}$.

Armed with Lemma 2, one can obtain that Proposition 2-(ii) is sufficient for optimality of a binary signal using the following argument. Suppose (Q, F) is a monopolist optimal outcome in which Q is F-ICC, and p_Q has at most one jump. Consider two cases. The first case is when p_Q is constant. In this case, concavity of $\hat{\pi}_{\underline{\theta}}$ implies that π_Q is concave as well. It follows it is without loss for F to have a singleton support: since π_Q is concave, pooling all signal realizations together into one realization weakly increases the value of the information design program (7). Consider now the second case in which p_Q has one jump at some θ^*. Again, because $\hat{\pi}_{\underline{\theta}}$ is concave (and marginal quality costs are constant), the function π_Q must be concave when restricted to $[\underline{\theta}_F, \theta^*]$ or $[\theta^*, \bar{\theta}_F]$. Consequently, pooling all realizations in $[\underline{\theta}_F, \theta^*]$ into one signal realization, and all realizations in $[\underline{\theta}^*, \bar{\theta}_F]$ into another signal realizations must weakly increase the value of information design program (7). In other words, one can replace any non-binary F with another incentive compatible signal whose support is binary without decreasing the monopolist's profit. That is, a binary signal is optimal.

Proposition 2 has the following implication: under the proposition's conditions, there is a monopolist optimal menu that contains no more than two purchasing options. For an explanation, note that restricting the menu to include only those alternatives that the buyer chooses with positive probability has no impact on the buyer's decisions as to what to learn and what to buy. Consequently, the monopolist can always maximize profits with a menu that contains weakly less options than the number of realizations in the support of the buyer's signal. Hence, if the buyer's signal is binary, the monopolist's menu can be binary as well.

Next, we discuss when the monopolist finds it optimal to offer the buyer only one purchasing option. Formally, an outcome (Q, F) is a **single-quality outcome** if $Q(\text{supp}(F))$ has only one non-zero element. Our next result characterizes which incentive compatible outcomes features a single (non-zero) quality. We also provide sufficient conditions for such outcomes to be optimal.

Corollary 4. *An IC outcome (Q, F) is a single-quality outcome if and only if $\text{supp}(F) \setminus \{\underline{\theta}_Q\}$ is a singleton. Moreover, a single-quality outcome is monopolist optimal whenever learning costs satisfy the high steepness condition, and one of the following two conditions hold:*

(i) If $\hat{\pi}_{\underline{\theta}^*}$ is concave for every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, a monopolist optimal outcome (Q, F) exists such that F is uninformative. Consequently, (Q, F) is a single-quality outcome.

(ii) If $\hat{\pi}_{\underline{\theta}^*}$ is convex for every $\underline{\theta}^* \in [\underline{\theta}, \theta_0]$, a monopolist optimal outcome (Q, F) exists such that $\mathrm{supp}(F) = \{\underline{\theta}_Q, \bar{\theta}_Q\}$. Consequently, (Q, F) is a single-quality outcome.

The proof of Corollary 4 is straightforward given our previous results. We first explain the characterization of which IC outcomes feature a single-quality. This characterization states that a single-quality outcomes must feature a signal structure with at most two realizations, exactly one of which obtains a strictly positive quantity. Sufficiency of these conditions follows similar lines to our binary-menu discussion. That these conditions are necessary follows from our ability to replace Q with an F-ICC allocation, along with the observation that F-ICC allocations must strictly increase over $[\underline{\theta}_F, \bar{\theta}_F]$.

We now explain the reasoning behind Corollary 4's sufficient conditions for optimality of a single-quality outcome. As discussed above, when learning costs are steep, all IC outcomes involve signals that do not separate any types—that is, (Q, F) is IC only if $I_F(\theta) > 0$ for all $\theta \in (\underline{\theta}, \bar{\theta})$. Thus, if Q were F-ICC, its induced marginal price would be constant. Therefore, the monopolist per-realization expected profit, π_Q, would be equal to $\hat{\pi}_{\underline{\theta}_Q}$ over the interval $[\underline{\theta}_Q, \bar{\theta}_Q]$. The desired result then follows from the information design problem in Corollary 1. Specifically, whenever the conditions of Corollary 4's part (i) hold, the program (7) admits an uninformative solution, whereas the conditions of Corollary 4's part (ii) imply (7) admits a binary solution supported on $\{\underline{\theta}_Q, \bar{\theta}_Q\}$. In both cases, one gets a single-quality monopolist-optimal outcome—provided it is optimal for the monopolist to serve some consumers, which is guaranteed by our assumption that $\kappa'(0) < \theta_0$.

Corollary 4 can be seen as a generalization of Theorem 3 in Mensch (2022). Within our model, one can interpret that theorem as providing sufficient conditions for a single-quality menu to be optimal when marginal quality costs are fixed at zero and the state is binary. Our argument is similar to that of Mensch (2022) in that it exploits the curvature of the seller's objective. Thus, one can interpret Corollary 4 as showing that Mensch's (2022) theorem applies in mean-measurable settings, so long as learning costs are sufficiently steep.

Corollary 4's sufficient conditions are easier to check when marginal quality costs are constant—i.e., when $\kappa(q) = \kappa_0 q$ for some $\kappa_0 > 0$. The reason is that, in this case, the derivative of $\hat{\pi}_{\underline{\theta}^*}$ equals

$$\hat{\pi}'_{\underline{\theta}^*}(\theta) = (\theta - \kappa_0)c''(\theta),$$

which is an expression that does not depend on $\underline{\theta}^*$. Therefore, whether $\hat{\pi}_{\underline{\theta}^*}$ is convex or concave $\underline{\theta}^*$ depends only on whether the above expression is increasing or decreasing in θ.

Comparing Corollary 4 to the standard analysis with a fully-informed buyer is instructive. In that setup, the monopolist typically finds it optimal to offer a single-purchasing option in one of two fairly restrictive cases: either marginal quality costs are constant, (e.g., Riley and Zeckhauser, 1983; Manelli and Vincent, 2007), or the marginal profit of serving a positive quality to any type other than the highest is negative, in which case the monopolist chooses to only serve the highest type. Corollary 4 is analogous to the second case, but where the type distribution is endogenous to the menu. As Corollary 4 points out, in our setting a single-quality menu is optimal whenever the monopolist wants to either deter information acquisition, or induce the buyer to acquire a binary signal structure whose lowest type is excluded. We thereby provide a new set of predictions, showing that single-quality menus can arise for general type distributions even if marginal quality costs are not constant.

6. Concluding Remarks

We conclude our paper with a few brief remarks regarding our assumptions and results.

Support vs. positive probability. Corollary 2 shows that, whenever learning costs are not too flat and marginal costs of quality are strictly increasing, the monopolist distorts downward the quality she provides to *all* buyer types, including the one with maximal valuation. This result stands in contrast to the conclusion one obtains when information is exogenous, where highest buyer type is allocated the efficient quality. As such, our paper suggests that an analyst who examines the market under the assumption that information is exogenous may come to erroneous conclusions regarding the efficiency of the market's allocation. However, one might wonder whether this error actually occurs: since the buyer's type distribution is endogenous, the buyer may choose an F that assigns zero probability to the top of its support. It turns out, however, that, under the corollary's assumption, it is without loss for F to put positive probability on $\bar{\theta}_F$. This observation follows from Proposition 1, which implies it is without loss to require that F is a bipooling. For an explanation, note that the corollary's assumption that $c'(\bar{\theta}) - c'(\theta_0) > \bar{q}$ implies that $\bar{\theta}_F < \bar{\theta}$. This means that, the top of the support of F must be in a bi-pooling interval—i.e., an interval over which F has a most two signal realizations. It follows that, whenever the Corollary's assumptions hold, and whenever F_0 is a full-support continuous distribution, it is without loss for F to assign positive probability to the top of its support.

Inefficiency of serving some types. Throughout, we assumed that $\kappa'(0) < \underline{\theta}$, meaning it is efficient to serve all types. We use this assumption to prove Theorem 3, Corollary 3, and Corollary 4. In Theorem 3, we use the assumption that $\kappa'(0) < \underline{\theta}$ to show that excluding $\underline{\theta}_F$ is inefficient. Thus, the theorem continues to hold whenever it is efficient to serve all types that can arise in a signal that is incentive compatible for some menu. More specifically, suppose there is some $\hat{\underline{\theta}}$ such that $c'(\hat{\underline{\theta}}) = c'(\theta_0) - \bar{q}$. Then Theorem 3 holds as stated so long $\kappa'(0) < \hat{\underline{\theta}}$. The reason is simple: by

Theorem 2, a signal F is IC for some allocation only if $\underline{\theta}_F \geq \hat{\underline{\theta}}$.[21] Therefore, it is always optimal to serve the lowest realized buyer type. The rest of the theorem's proof goes through as before.

The assumption that $\kappa'(0) < \hat{\underline{\theta}}$ is also sufficient for extending Corollary 3 and Corollary 4. Corollary 4 uses the assumption that $\kappa'(0) < \underline{\theta}$ to argue that serving some types is always profitable for the monopolist. The same holds if $\kappa'(0) < \hat{\underline{\theta}}$. As for Corollary 3, it uses the assumption that $\kappa'(0) < \underline{\theta}$ to guarantee that the monopolist can strictly benefit from uniformly increasing the quality it provides to all type realizations according to the buyer's chosen signal. One can show $\kappa'(0) < \hat{\underline{\theta}}$ guarantees the same benefit, and so is sufficient for the corollary's proof to go through.[22]

What happens if none of the above-mentioned replacements holds? We know that Corollary 3 may fail: there are examples where $\kappa(q) = \kappa_0 q$, $\kappa_0 \geq \theta_0$, and the monopolist optimal outcome has $Q(\bar{\theta}_F) < \bar{q}$. Corollary 4 would also fail, but only because it may be optimal for the monopolist to not serve anyone at the optimum. Whenever the monopolist optimal outcome results in the buyer being served with positive probability, the proposition's sufficient conditions for a single-quality outcome still hold.

On the other hand, Theorem 3 always holds, regardless of whether it is efficient to serve the lowest possible type. To show this, one can show there is at most one signal realization $\theta^* \in \operatorname{supp}(F)$ for which $\kappa'(Q(\theta^*)) > \theta^*$. Moreover, this realization must be excluded, meaning that $\theta^* = \underline{\theta}_Q$. So, in effect, the monopolist never provides an inefficiently high, positive quality to any realized θ. We provide a proof sketch in the online appendix.

Affine Learning Costs. We assume the buyer's costs of learning were affine in the distribution of her posterior estimate. As explained earlier, the assumption that costs depend only on the distribution of the buyer's posterior estimate is without loss. However, the assumption that costs are affine in this distribution is substantive. Next, we explain how our analysis changes if we instead assumed the buyer's learning costs can be locally approximated by an affine function. Specifically, say C is **Gateaux differentiable** if every $F \in \mathcal{I}$ admits some twice differentiable strictly convex function $c_F : \Theta \to \mathbb{R}$ such that for every $G \in \mathcal{I}$,

$$\lim_{\epsilon \searrow 0} \frac{1}{\epsilon} \left[C(F + \epsilon(G - F)) - C(F) \right] = \int c_F(\theta)(G - F)(\mathrm{d}\theta).$$

Assumptions of this type were first introduced into information acquisition models by Ravid, Roesler, and Szentes (2022). Lipnowski and Ravid (2022) introduce a generalization of this class of cost functions that are not mean-measurable—i.e., costs that depend on the distribution of the buyer's posterior *belief*.

[21]Formally, if F is IC for some allocation, it is also IC for some F-ICC allocation Q. Consequently, $\bar{q} \geq Q(\bar{\theta}_F) - Q(\underline{\theta}) \geq c'(\bar{\theta}_F) - c'(\underline{\theta}_F) \geq c'(\theta_0) - c'(\underline{\theta}_F)$. That $\underline{\theta}_F \geq \hat{\underline{\theta}}$ then follows from c' being strictly increasing.

[22]The argument remains exactly the same, except that now one obtains a contradiction by noting that (15) holding for $\theta^* = \underline{\theta}_F$ implies that $\kappa_0 > \underline{\theta}_F \geq \hat{\underline{\theta}}_F$.

Under Gateaux differentiability, our reduction to F-ICC mechanisms is still without loss, though one must adjust the definition of ICC allocations so that $c_F^{'}$ replaces c'. To get this result, one first applies Lemma 1 from Georgiadis, Ravid, and Szentes (2022) to get that F solves the buyers problem if and only if it solves the buyer's problem when her costs are given by their affine approximation at F.[23] Said differently, a given F is incentive compatible for the buyer if and only if it is buyer optimal when the buyer's cost function is given by $G \mapsto \int c_F(\theta)G(\mathrm{d}\theta)$. Theorem 2 therefore implies it is without loss to focus on (the properly defined) F-ICC mechanisms.

The appropriate adjustment of Corollary 1 also continues to hold. In particular, it turns out the adjusted mechanism design problem 6 is amenable to the same techniques as the problem presented in this paper. Consequently, Lemma 8 and Corollary 3 remain valid as stated. The reason is that these results rely only on perturbations of the F-marginal price. Lemma 2 also holds as stated, since it involves keeping information constant.

The rest of our results, however, cannot be proven as is. The reason is that the affine approximation depends on F, making the analogue of the information design problem (7) much less tractable. For an explanation, note that we prove our other results by fixing an ICC allocation Q and varying F within the set of signals for which Q is ICC. When learning costs are affine, this set contains any signal that separates any θ at which p_Q is strictly increasing. The same is not true when C is Gateaux differentiable: in that case, changing F typically involves changing c_F', and therefore the allocation. Hence, the arguments for the rest of our results do not apply when costs are merely Gateaux differentiable.

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[23]The lemma is stated for the case where the agent can induce any distribution, but applies as stated to any convex constraint set.

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A. Online Proofs Appendix

A.1. Cost Function Characterization

In this section, we show a continuous cost function $C : \mathcal{I} \to \mathbb{R}$ is affine and strictly increasing in informativeness if and only if a strictly convex continuous function $c : \Theta \to \mathbb{R}$ exists such that $C(F) = \int c(\theta)F(\mathrm{d}\theta)$. To prove this result, note the Riesz representation theorem implies C is continuous and affine if and only if $C(F) = \int \tilde{c}(\theta)F(\mathrm{d}\theta)$ for some continuous $\tilde{c} : \Theta \to \mathbb{R}$. All that remains is to show $\tilde{c}$ must be strictly convex. For this purpose, fix any $x, y, z \in \left(\underline{\theta}, \bar{\theta}\right)$ such that $y = \beta x + (1-\beta) z$ for some $\beta \in (0,1)$. By Lemma 6 in Ravid, Roesler, and Szentes (2020), one can find $F', F'' \in \mathcal{I}$ and $\gamma > 0$ such that $F' \succ F''$, and

$$F' - F'' = \gamma \left(\beta \mathbf{1}_{\left[x,\bar{\theta}\right]} + (1-\beta) \mathbf{1}_{\left[z,\bar{\theta}\right]} - \mathbf{1}_{\left[y,\bar{\theta}\right]} \right),$$

where for any $w \in \Theta$, $\mathbf{1}_{[w,\bar{\theta}]}$ is the CDF of the distribution that generates w with probability 1. Since C is strictly increasing in $\succeq$, it follows that

$$0 < C(F') - C(F'') = \int c(\theta)(F' - F'')(\mathrm{d}\theta) = \gamma \left(\beta \tilde{c}(x) + (1-\beta) \tilde{c}(z) - \tilde{c}(y) \right).$$

The claim follows.

A.2. Proof of Theorem 1

We begin by formally defining the buyer's maximization problem holding the monopolist's menu fixed. Let $X = [0, \bar{q}] \times \mathbb{R}_+$, and endow the set of Borel measures over $X \times \Theta$, $\Delta(X \times \Theta)$, with the weak* topology. Given a menu M, the buyer's program can be written as

$$\begin{aligned} \max_{\xi \in \Delta(X \times \Theta)} & \int (\theta q - t)\, \xi\left(\mathrm{d}(q,t,\theta)\right) - C\left(\mathrm{marg}_{\Theta}\, \mu\right) \\ \text{s.t.} & \operatorname{supp} \mu \subseteq M \times \Theta, \\ & \mathrm{marg}_{\Theta} \mu \preceq F_0. \end{aligned}$$

Observe the above program involves the maximization of a continuous objective over a compact constraint set, and so the set of solution, $\Xi(M)$, is non-empty for every compact M. Letting $\mathcal{M}$ be the collection of compact subsets of X that contain the non-participation option $(0,0)$, the monopolist's program can be written as

$$\max_{(M,\xi)\in\mathcal{M}\times\Delta(X\times\Theta)}\int\left(t-\kappa\left(q\right)\right)\xi\left(\mathrm{d}(q,t,\theta)\right)$$
$$\text{s.t.}\ \xi\in\Xi\left(M\right).$$

Notice it is without loss to assume $M\subseteq\bar{X}=[0,\bar{q}]\times\left[0,\bar{\theta}\bar{q}\right]$, because the buyer strictly prefers $(0,0)$ to any menu item that includes a transfer strictly above $\bar{\theta}\bar{q}$. Let $\mathcal{K}\left(\bar{X}\right)$ be the set of all compact non-empty subsets of $\bar{X}$ endowed with the Hausdorff metric,

$$d\left(A,B\right)=\max\left\{\max_{b\in B}\min_{a\in A}d\left(b,a\right),\max_{a\in A}\min_{b\in B}d\left(a,b\right)\right\},$$

and take $\bar{\mathcal{M}}$ to be the elements of $\mathcal{K}\left(\bar{X}\right)$ that contain $(0,0)$. Taking $\bar{\Xi}$ to be the restriction of Ξ to $\bar{\mathcal{M}}$, and letting

$$\mathrm{Gr}\ \bar{\Xi}=\left\{(M,\xi)\in\bar{\mathcal{M}}\times\Delta(X\times\Theta):\xi\in\bar{\Xi}(M)\right\}$$

denote the restriction's graph, we get that the monopolist's problem can be rewritten as

$$\max_{(M,\xi)\in\mathrm{Gr}\ \bar{\Xi}}\int\left(t-\kappa\left(q\right)\right)\xi\left(\mathrm{d}(q,t,\theta)\right). \tag{8}$$

Observe $\bar{\mathcal{M}}$ is a closed subset of $\mathcal{K}\left(\bar{X}\right)$, and so because $\mathcal{K}\left(\bar{X}\right)$ is compact (Aliprantis and Border (2006), Theorem 3.85), $\bar{\mathcal{M}}$ must be compact as well. It follows, by Berge's theorem of the maximum, that $\bar{\Xi}$ is upper-hemicontinuous and has a closed graph (Aliprantis and Border (2006), Theorem 17.10). Hence, this graph must be compact because it is a subset of $\bar{\mathcal{M}}\times\Delta\left(\bar{X}\times\Theta\right)$, which is compact. That (8) admits a solution follows.

A.3. Jumps Towards Efficiency

In this section we prove that jumps towards efficiency are without loss of optimality. Moreover, we show that whenever Q jumps towards efficiency, $\pi_{Q,\underline{u}}$ is upper-semicontinuous.

We begin with proving that jumping towards efficiency is without loss of optimality.

Lemma 3. *Suppose $(Q,\underline{u},F)$ is IC. Then there is an allocation Q^* that jumps towards efficiency such that $(Q^*,\underline{u},F)$ is IC, and $\int\pi_{Q^*,\underline{u}}(\theta)F(\mathrm{d}\theta)\geq\int\pi_{Q,\underline{u}}(\theta)F(\mathrm{d}\theta)$. Moreover, if $(Q,\underline{u},F)$ is monopolist optimal and κ is strictly convex, Q^* equals Q F-almost surely.*

Proof of Lemma 3. Suppose Q is F-IC. Define the allocation Q^* via

$$Q^*(\theta):=\min\left\{\tilde{q}:\tilde{q}\in\operatorname*{argmax}_{q\in[Q_-(\theta),Q_+(\theta)]}\theta q-\kappa\left(q\right)\right\},$$

which is well-defined because κ is convex. Note that Q^* equals Q at any θ where Q is continuous. Since Q is discontinuous in at most a countable set of points, we get that $Q*_+ = Q_+$ and $Q*_- = Q_-$. It follows Q jumps towards efficiency. In addition, note that Q and Q^* differ on at most a countable set of points, which has Lebesgue measure zero, and so $V_{Q,\underline{u}} = V_{Q^*,\underline{u}}$. It follows $(Q, \underline{u}, F)$ is IC if and only if $(Q^*, \underline{u}, F)$ is IC as well.

We now turn to showing the monopolists profit under $(Q^*, \underline{u}, F)$ is no lower than it is under $(Q, \underline{u}, F)$. To see this, note that

$$\int [\pi_Q(\theta) - \pi_{Q^*}(\theta)]F(\mathrm{d}\theta) = \int [(\theta Q(\theta) - \kappa(Q(\theta))) - (\theta Q^*(\theta) - \kappa(Q^*(\theta)))] F(\mathrm{d}\theta) \leq 0, \quad (9)$$

where the first equality follows from $V_{Q,\underline{u}} = V_{Q^*,\underline{u}}$.

We now conclude the proof by arguing that if (Q, F) is monopolist optimal, and κ is strictly convex, then $Q^* = Q$ F-almost surely. For this purpose, note that strict convexity of κ means that Q^* is the unique maximizer of $q \mapsto \theta[q - \kappa(q)]$ over $[Q_-(\theta), Q_+(\theta)]$. Consequently, (9) must hold with strict inequality whenever $Q^* \neq Q$ over a positive F-measure set. But $(Q^*, \underline{u})$ is F-IC, and so $(Q, \underline{u}, F)$ being monopolist optimal means that the inequality in (9) must hold with equality. It follows that $Q^* = Q$ F-almost surely. □

Next, we show that Q jumping towards efficiency has a useful technical benefit: it makes π_Q upper semicontinuous.

Lemma 4. *If an allocation Q jumps towards efficiency, π_Q is upper semicontinuous.*

Proof of Lemma 4. Suppose Q jumps towards efficiency. In what follows, define

$$Q^e_-(\theta) = \min\{q : q \in \operatorname*{argmax}_{\tilde{q}\in[0,\bar{q}]}[\theta\tilde{q} - \kappa(\tilde{q})]\}, \quad \text{and} \quad Q^e_+(\theta) = \max\{q : q \in \operatorname*{argmax}_{\tilde{q}\in[0,\bar{q}]}[\theta\tilde{q} - \kappa(\tilde{q})]\}.$$

It is easy to verify that both Q^e_- and Q^e_+ are increasing. Berge's Maximum Theorem then delivers that Q^e_- and Q^e_+ are respectively lower and upper semicontinuous, which (combined with monotonicity of the two functions) delivers that the functions are respectively left and right continuous. Moreover, Berge's Maximum Theorem also delivers that $\lim_{\theta_n \nearrow \theta} Q^e_+(\theta) = Q^e_-(\theta)$ and $\lim_{\theta_n \searrow \theta} Q^e_-(\theta) = Q^e_+(\theta)$. Finally, $\operatorname{argmax}_{q\in[0,\bar{q}]} \theta q - \kappa(q)) = [Q^e_-(\theta), Q^e_+(\theta)]$ holds because $q \mapsto (\theta q - \kappa(q))$ is a concave function.

Let $(\theta_n)_{n\in\mathbb{N}}$ be some convergent sequence, and take θ_∞ to be its limit. Our goal is to show that $\limsup_n \pi_Q(\theta_n) \leq \pi_Q(\theta_\infty)$. Since V_Q is continuous, it is sufficient to show that

$$\limsup_n [\theta_n Q(\theta_n) - \kappa \circ Q(\theta_n)] \leq \theta_\infty Q(\theta_\infty) - \kappa \circ Q(\theta_\infty). \quad (10)$$

Obviously, equation (10) holds if Q is continuous at θ_∞. Thus, from now on we assume a discontinuity in Q at θ_∞, so $Q_-(\theta_\infty) < Q_+(\theta_\infty)$. Moreover, since every sequence admits a monotone subsequence, to show the above inequality it is sufficient to show that it holds when $(\theta_n)_{n\in\mathbb{N}}$ is monotone. Without loss of generality, suppose θ_n is monotone increasing.

We proceed in cases.

Case 1. Suppose that $Q(\theta_\infty) \in [Q^e_-(\theta), Q^e_+(\theta)]$ (i.e., $Q(\theta_\infty)$ is efficient). Then for every n,

$$\begin{aligned}\theta_n Q(\theta_n) - \kappa\circ Q(\theta_n) \leq \theta_n Q^e_+(\theta_n) - \kappa\circ Q^e_+(\theta_n) &\to \theta_\infty Q^e_-(\theta_\infty) - \kappa\circ Q^e_-(\theta_\infty)\\ &= \theta_\infty Q(\theta_\infty) - \kappa\circ Q(\theta_\infty),\end{aligned}$$

where convergence follows from Berge's Maximum Theorem.

Case 2. Suppose $Q(\theta_\infty) < Q^e_-(\theta)$. Because $q \mapsto \theta q - \kappa(q)$ is concave and Q jumps towards efficiency, $Q(\theta_\infty) = Q_+(\theta_\infty) < Q^e_-(\theta_\infty)$. In the next paragraph we argue that $Q(\theta_n) < Q^e_-(\theta_n)$ holds for all sufficiently large n. Taking this inequality as given, note that, because $\theta_n \nearrow \theta_\infty$, $Q^e_-(\theta_n) \to Q^e_-(\theta_\infty)$, and so for all sufficiently large n, $Q(\theta_n) \leq Q(\theta_\infty) < Q^e_-(\theta_n)$. Since $q \mapsto \theta q - \kappa(q)$ is concave, we get that

$$\theta_n Q(\theta_n) - \kappa\circ Q(\theta_n) \leq \theta_n Q(\theta_\infty) - \kappa\circ Q(\theta_\infty) \to \theta_\infty Q(\theta_\infty) - \kappa\circ Q(\theta_\infty),$$

as required.

Thus, to complete the proof of this case, it remains to show that $Q(\theta_n) < Q^e_-(\theta_n)$ holds for all sufficiently large n. Suppose otherwise. Then one can find a subsequence $(Q_n)_{n\in N}$ for some infinite $N \subseteq \mathbb{N}$ such that $Q(\theta_n) \geq Q^e_-(\theta_n)$ for all $n \in N$. Since Q^e_- is left continuous and $\theta_n \nearrow \theta_\infty$, one gets that

$$Q(\theta_\infty) \geq Q(\theta_n) \geq Q^e_-(\theta_n) \to Q^e_-(\theta_\infty),$$

meaning $Q(\theta_\infty) \geq Q^e_-(\theta_\infty)$, a contradiction. The proof of this case is therefore complete.

Case 3. Suppose $Q(\theta_\infty) > Q^e_+(\theta)$. Then concavity of $q \mapsto \theta q - \kappa(q)$ and Q jumping towards efficiency means that $Q(\theta_\infty) = Q_-(\theta_\infty)$. Since $\theta_n \nearrow \theta_\infty$ and Q is monotone, $Q(\theta_n) \to Q_-(\theta_\infty) = Q(\theta_\infty)$. It follows (10) holds.

□

A.4. Proofs from Section 3

We begin with stating Dizdar and Kováč's (2020) generalization of Dworczak and Martini's (2019) duality result. Towards this goal, for every $F \in \mathcal{I}$, define an F-**price** to be a Lipschitz continuous, convex function that is affine on any interval of F-pooling types, that is, over any interval $\left(\underline{\theta}_0, \bar{\theta}_0\right) \subseteq \{\theta : I_F(\theta) > 0\}$. The following result relates this concept to the buyer's problem.

Theorem 4 (Dizdar and Kováč, 2020)**.** *Let $\Phi : \Theta \to \mathbb{R}$ be a bounded, upper semicontinuous function that admits some $\epsilon > 0$ and $\bar{L} \in \mathbb{R}_+$ such that $\Phi(\theta) - \Phi(\theta') \geq \bar{L}(\theta' - \theta)$ holds for $\underline{\theta} \leq \theta' \leq \theta \leq \underline{\theta} + \epsilon$, and $\Phi(\theta) - \Phi(\theta') \leq \bar{L}(\theta - \theta')$ holds for $\bar{\theta} - \epsilon \leq \theta' \leq \theta \leq \bar{\theta}$. Then,*

$$F \in \operatorname{argmax} \int \Phi(\theta) F(\mathrm{d}\theta)$$

if and only if an F-price P exists such that $P(\theta) \geq \Phi(\theta)$ for θ, where the inequality holds with equality for $\theta \in \operatorname{supp} F$.[24]

We now use this theorem to prove Lemma 1.

Proof of Lemma 1. We begin by arguing that $N_{Q,\underline{u}}$ is satisfies the pre-requisites of Theorem 4. By definition, $N_{Q,\underline{u}}$ is continuous, and therefore bounded and upper-semicontinuous. Now pick any $\epsilon > 0$. Then for any $\theta' < \theta$ such that $(\theta', \theta) \in [\underline{\theta}, \underline{\theta} + \epsilon]$ we have

$$N_{Q,\underline{u}}(\theta) - N_{Q,\underline{u}}(\theta') = \int_{\tilde{\theta} \in (\theta', \theta)} \left[Q(\tilde{\theta}) - c'(\tilde{(\theta)})\right] \mathrm{d}\tilde{\theta} \geq -c'(\underline{\theta} + \epsilon)(\theta - \theta') = c'(\underline{\theta} + \epsilon)(\theta' - \theta),$$

where the inequality follows from $Q \geq 0$ and c' being increasing. Similarly, for every $\theta' < \theta$ such that $(\theta', \theta) \in [\bar{\theta} - \epsilon, \bar{\theta}]$,

$$N_{Q,\underline{u}}(\theta) - N_{Q,\underline{u}}(\theta') = \int_{\tilde{\theta} \in (\theta', \theta)} \left[Q(\tilde{\theta}) - c'(\tilde{(\theta)})\right] \mathrm{d}\tilde{\theta} \leq (\bar{q} - c'(\bar{\theta} - \epsilon))(\theta - \theta'),$$

where the inequality follows from $Q \leq \bar{q}$ and c' being increasing. It follows $N_{Q,\underline{u}}$ satisfies the Theorem 4 assumptions about Φ.

We now turn to proving the "if" part of Lemma 1. For this part, note that the Lemma's conditions imply $P_{Q,\underline{u},p}$ is an F-price. Theorem 4 then delivers that F is buyer optimal.

For the "only if" part, suppose F solves the buyer's problem. Let P be the F-price delivered by Theorem 4. Since P is Lipschitz, there is a function $p : \Theta \to \mathbb{R}$ such that $P(\theta) = P(\underline{\theta}_F) + \int_{\underline{\theta}_F}^{\theta} p(\tilde{\theta})\mathrm{d}\tilde{\theta}$ for all θ. Notice that $P(\underline{\theta}_F) = N_{Q,\underline{u}}(\underline{\theta}_F)$, and so $P = P_{Q,\underline{u},p}$. Therefore, we have and $P_{Q,\underline{u},p} = P \geq N_{Q,\underline{u}}$, with the inequality holding with equality over the support of F.

[24]The statement of the theorem here is slightly more general than the one stated by Dizdar and Kováč (2020). However, the exact same steps as in their proof hold, with $\bar{L}$ replacing L in equation (8) of their paper.

Next, we claim one can take p such that $p(\theta) = Q(\theta) - c'(\theta)$ for all $\theta \in \text{supp } F$. To do so, we show below that for each such θ,

$$p_+(\theta) \geq Q_+(\theta) - c'(\theta) \geq Q_-(\theta) - c'(\theta) \geq p_-(\theta). \tag{11}$$

Equation 11 immediately implies the desired equality whenever $p_+(\theta) = p_-(\theta)$. Moreover, since P is convex, we can take p to be increasing, and so $p_+(\theta) > p_-(\theta)$ holds over a Lebesgue-null set, meaning one can edit p so that it satisfies the desired equality without impacting its integral. Hence, to prove that one can take p such that $p(\theta) = Q(\theta) - c'(\theta)$ for all $\theta \in \text{supp } F$, it is sufficient to show that (11) holds for all such θ. To show this inequality, notice that for any $\epsilon > 0$,

$$\begin{aligned}
0 &\leq \frac{1}{\epsilon}\Big[P(\theta-\epsilon) - N_{Q,\underline{u}}(\theta-\epsilon)\Big] \\
&= \frac{1}{\epsilon}\Big[P(\theta-\epsilon) - P(\theta) + N_{Q,\underline{u}}(\theta) - N_{Q,\underline{u}}(\theta-\epsilon)\Big] \\
&\xrightarrow{\epsilon\searrow 0} -p_-(\theta) + Q_-(\theta) - c'(\theta),
\end{aligned}$$

where the first inequality follows from $P \geq N_{Q,\underline{u}}$ and the second equality from P being equal to $N_{Q,\underline{u}}$ over the support of F (and θ being in that support). It follows $Q_-(\theta) - c'(\theta) \geq p_-(\theta)$. An analogous argument delivers that $p_+(\theta) \geq Q_+(\theta) - c'(\theta)$.

Thus, all that remains is to argue that we can take p so that it satisfies the conditions of an F-price. Towards this goal, recall we already argued we can take p to be increasing. Moreover, because P is affine on any interval over which I_F is strictly positive, p is constant over any such interval. Continuity of I_F then implies that if $I_F(\theta)$ is strictly positive, I_F must be strictly positive over a neighborhood of θ, and so p must be constant around θ.

Finally, we need to argue that $p \in [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$. To do so, notice first it is without loss to choose p so that $p(\bar{\theta}) = p_-(\bar{\theta}_F)$: either $\bar{\theta}_F = \bar{\theta}$, or $\bar{\theta}_F < \bar{\theta}$, in which case I_F must be strictly positive over $(\bar{\theta}_F - \epsilon, \bar{\theta})$ for some small $\epsilon > 0$, meaning p must constant over the same, and so setting $p(\bar{\theta}) = p(\bar{\theta}_F) = p_-(\bar{\theta}_F)$ is without loss. An analogous argument delivers it is without loss to have $p(\underline{\theta}) = p_+(\underline{\theta}_F)$. Combined with equation 11, we get that

$$p(\bar{\theta}) = p(\bar{\theta}_F) = Q(\bar{\theta}_F) - c(\bar{\theta}_F) \leq \bar{q} - c(\bar{\theta}_F).$$

An analogous argument delivers $p(\underline{\theta}) = p(\underline{\theta}_F) \geq -c'(\underline{\theta}_F)$. Since p is increasing, we get $p \in [-c'(\underline{\theta}_F), \bar{q} - c'(\bar{\theta}_F)]$ as required. □

Finally, we prove that focusing on F-ICC mechanisms is without loss of generality.

Proof of Theorem 2. As a preliminary step, suppose F is IC for some mechanism, and that we have some F-ICC allocation $\hat{Q}$. Let $\hat{p} = p_Q$ be the F-marginal price associated with $\hat{Q}$. In the case where $\lim_{\theta \searrow \underline{\theta}} c'(\theta) = -\infty$ and $\lim_{\theta \nearrow \bar{\theta}} c'(\theta) = \infty$, because c' is continuous and strictly increasing, $\underline{\theta}_{\hat{Q}} \in (\underline{\theta}, \underline{\theta}_F]$ is the unique solution to $\hat{p}\left(\underline{\theta}_F\right) + c'\left(\underline{\theta}_{\hat{Q}}\right) = 0$, and $\bar{\theta}_{\hat{Q}} \in [\bar{\theta}_F, \bar{\theta})$ is the unique solution to $\hat{p}(\bar{\theta}_F) + c'(\bar{\theta}_{\hat{Q}}) = \bar{q}$. If the slope of c at the boundary is finite and no such respective solution exists in the interval $[\underline{\theta}, \bar{\theta}]$, we set $\underline{\theta}_{\hat{Q}} = \underline{\theta}$ and $\bar{\theta}_{\hat{Q}} = \bar{\theta}$, respectively.

We first show every F-ICC mechanism is F-IC. Let Q be an F-ICC allocation, take $p := p_Q$. Consider the F-price $P_{Q,\underline{u},p}$ as defined in equation (4). By Lemma 1, to show that $(Q, \underline{u})$ is F-IC, showing that $P_{Q,\underline{u},p}(\theta) \geq N_{Q,\underline{u}}(\theta)$ holds for all θ, and that $P_{Q,\underline{u},p}(\theta) = N_{Q,\underline{u}}(\theta)$ for all $\theta \in \operatorname{supp} F$ suffices. We begin by showing that $P_{Q,\underline{u},p}(\theta) = N_{Q,\underline{u}}(\theta)$ holds for all $\theta \in [\underline{\theta}_Q, \bar{\theta}_Q] \supseteq [\underline{\theta}_F, \bar{\theta}_F]$. Indeed, for each such θ,

$$N_{Q,\underline{u}}(\theta) = N_{Q,\underline{u}}(\underline{\theta}_F) + \int_{\underline{\theta}_F}^{\theta} (Q(\tilde{\theta}) - c'(\tilde{\theta}))\mathrm{d}\tilde{\theta} = N_{Q,\underline{u}}(\underline{\theta}_F) + \int_{\underline{\theta}_F}^{\theta} p(\tilde{\theta})\mathrm{d}\tilde{\theta} = P_{Q,\underline{u},p}(\theta),$$

as required. It remains to show $P_{Q,\underline{u}} \geq N_{Q,\underline{u}}$ for all $\theta \in \Theta \setminus \left[\underline{\theta}_Q, \bar{\theta}_Q\right]$. We show this inequality for $\theta \in [\underline{\theta}, \underline{\theta}_Q)$, with the argument for $\theta \in (\bar{\theta}_Q, \bar{\theta}]$ being analogous. Since p is an F-marginal price, we have $p(\underline{\theta}_F) + c'(\underline{\theta}_F) \leq \bar{q} - c'(\bar{\theta}_F) + c'(\underline{\theta}_F) \leq \bar{q}$. Therefore, since both p and c' are increasing, we get that for every $\theta < \underline{\theta}_Q$,

$$Q(\theta) = \max\{p(\theta) + c'(\theta), 0\} \geq p(\theta) + c'(\theta),$$

meaning $p(\theta) \leq Q(\theta) - c'(\theta)$. Therefore,

$$\begin{aligned}
P_{Q,\underline{u},p}(\theta) &= N_{Q,\underline{u}}\left(\underline{\theta}_F\right) + \int_{\underline{\theta}_F}^{\theta} p(\tilde{\theta})\mathrm{d}\tilde{\theta} = N_{Q,\underline{u}}\left(\underline{\theta}_F\right) - \int_{\theta}^{\underline{\theta}_F} p(\tilde{\theta})\mathrm{d}\tilde{\theta} \\
&\geq N_{Q,\underline{u}}\left(\underline{\theta}_F\right) - \int_{\theta}^{\underline{\theta}_F} \left[Q(\tilde{\theta}) - c'(\tilde{\theta})\right] \mathrm{d}\tilde{\theta} \\
&= V_{Q,\underline{u}}(\underline{\theta}_F) - c(\underline{\theta}_F) - \int_{\theta}^{\underline{\theta}_F} \left[Q(\tilde{\theta}) - c'(\tilde{\theta})\right] \mathrm{d}\tilde{\theta} \\
&= \underline{u} + \int_{\underline{\theta}}^{\underline{\theta}_F} Q(\tilde{\theta})\mathrm{d}\tilde{\theta} - \left[c(\underline{\theta}) + \int_{\underline{\theta}}^{\underline{\theta}_F} c'(\tilde{\theta})\mathrm{d}\tilde{\theta}\right] - \int_{\theta}^{\underline{\theta}_F} \left[Q(\tilde{\theta}) - c'(\tilde{\theta})\right] \mathrm{d}\tilde{\theta} \\
&= \underline{u} + \int_{\underline{\theta}}^{\theta} Q(\tilde{\theta})\mathrm{d}\tilde{\theta} - \left[c(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} c'(\tilde{\theta})\mathrm{d}\tilde{\theta}\right] \\
&= V_{Q,\underline{u}}(\theta) - c(\theta) = N_{Q,\underline{u}}(\theta),
\end{aligned}$$

as required. Theorem 4 implies $(Q_{\underline{u}})$ is F-IC.

Next, we argue every F-IC mechanism admits an equivalent F-ICC mechanism. Let $(\tilde{Q}, \tilde{\underline{u}})$ be an F-IC mechanism. Take p be the F-marginal price delivered from Lemma 1 that certifies that $(\tilde{Q}, \tilde{\underline{u}}, F)$ is IC, in which $p(\theta) = \tilde{Q}(\theta) - c'(\theta)$ for all $\theta \in \operatorname{supp} F$. Let Q be the F-ICC allocation generated by p, and set $\underline{u}$ according to

$$\underline{u} = N_{\tilde{Q},\underline{\tilde{u}}}(\underline{\theta}_F) - N_{Q,0}(\underline{\theta}_F).$$

By choice of $\underline{u}$, $N_{Q,\underline{u}}(\underline{\theta}_F) = N_{\tilde{Q},\underline{\tilde{u}}}(\underline{\theta}_F)$. Consequently, $P_{Q,\underline{u},p} = P_{\tilde{Q},\underline{\tilde{u}},p}$. Note that for every $\theta \in \operatorname{supp} F \subseteq [\underline{\theta}_F, \bar{\theta}_F]$, we have both $Q(\theta) = p(\theta) + c(\theta) = \tilde{Q}(\theta)$ (since $p(\theta) = \tilde{Q}(\theta) - c'(\theta)$). For such θ, we also have $N_{Q,\underline{u}}(\theta) = P_{Q,\underline{u},p}(\theta) = P_{\tilde{Q},\underline{\tilde{u}},p}(\theta) = N_{\tilde{Q},\underline{\tilde{u}}}(\theta)$ (because Q is F-ICC), and so $V_{Q,\underline{u}}(\theta) = N_{Q,\underline{u}}(\theta) + c(\theta) = N_{\tilde{Q},\underline{\tilde{u}}}(\theta) + c(\theta) = V_{\tilde{Q},\underline{\tilde{u}}}(\theta)$. It remains only to show that $\underline{u} \geq \underline{\tilde{u}}$, which follows from

$$\underline{u} = N_{Q,\underline{u}}(\underline{\theta}_Q) + c(\underline{\theta}_Q) = P_{Q,\underline{u},p}(\underline{\theta}_Q) + c(\underline{\theta}_Q) = P_{\tilde{Q},\underline{\tilde{u}},p}(\underline{\theta}_Q) + c(\underline{\theta}_Q) \geq N_{\tilde{Q},\underline{\tilde{u}}}(\underline{\theta}_Q) + c(\underline{\theta}_Q) \geq \underline{\tilde{u}},$$

where the first equality follows from the fact shown earlier that $P_{Q,\underline{u},p}(\theta) = N_{Q,\underline{u}}(\theta)$ for all $\theta \in [\underline{\theta}_Q, \bar{\theta}_Q]$. The theorem's proof is now complete. □

A.5. Proof of Theorem 3

The ultimate goal of this section is to prove Theorem 3. En-route, we prove several auxiliary results about the programs (6) and (7).

A.5.1. Allocation Pertubations

We first consider the program (6). We begin with two lemmas. The first lemma notes the set of allocations that are F-IC is convex. The second lemma uses this convexity to derive a necessary first order condition for an allocation to be part of a monopolist optimal allocation.

Lemma 5. *Suppose Q and $\tilde{Q}$ are both F-IC. Then, $(1-\beta)\,Q + \beta\tilde{Q}$ is also F-IC for all $\beta \in [0,1]$.*

Proof. Note that for any two allocations Q, $\tilde{Q}$, and any $\beta \in [0,1]$,

$$\begin{aligned} V_{(1-\beta)Q+\beta\tilde{Q}}(\theta) &= \int_{\underline{\theta}}^{\theta} \left((1-\beta)\,Q\left(\tilde{\theta}\right) + \beta\tilde{Q}\left(\tilde{\theta}\right)\right) \mathrm{d}\tilde{\theta} \\ &= (1-\beta)\int_{\underline{\theta}}^{\theta} Q\left(\tilde{\theta}\right) \mathrm{d}\tilde{\theta} + \beta\int_{\underline{\theta}}^{\theta} \tilde{Q}\left(\tilde{\theta}\right) \mathrm{d}\tilde{\theta} = (1-\beta)\,V_Q(\theta) + \beta V_{\tilde{Q}}(\theta). \end{aligned}$$

Therefore, if both Q, $\tilde{Q}$ are F-IC, one obtains the following inequality for all $\tilde{F}$:

$$\begin{aligned}\int \left(V_{(1-\beta)Q+\beta\tilde{Q}} - c\right)(\theta)F\left(\mathrm{d}\theta\right) &= (1-\beta)\int \left(V_Q - c\right)(\theta)F\left(\mathrm{d}\theta\right) + \beta\int\left(V_{\tilde{Q}} - c\right)(\theta)F\left(\mathrm{d}\theta\right) \\ &\geq (1-\beta)\int \left(V_Q - c\right)(\theta)\mathrm{d}\tilde{F}(\theta) + \beta\int\left(V_{\tilde{Q}} - c\right)(\theta)\mathrm{d}\tilde{F}(\theta) \\ &= \int \left(V_{(1-\beta)Q+\beta\tilde{Q}} - c\right)(\theta)\mathrm{d}\tilde{F}(\theta),\end{aligned}$$

meaning $(1-\beta)\,Q + \beta\tilde{Q}$ is also F-IC. □

Next, we obtain a necessary first-order condition for the monopolist's optimal outcome by perturbing the allocation while keeping the buyer's information fixed.

Lemma 6. *Let* (Q^*, F^*) *be monopolist optimal. Suppose* Q *also incentivizes* F^*. *Then,*

$$\int \left[\left(\theta - \kappa'\left(Q^*(\theta)\right)\right)\left(Q - Q^*\right)(\theta) - \left(V_Q - V_{Q^*}\right)(\theta)\right] F^*(\mathrm{d}\theta) \leq 0.$$

Proof. Suppose (Q^*, F^*) is monopolist optimal, and let Q be any other F^*-IC allocation. Defining the allocation $Q_\varepsilon := Q^* + \epsilon\left(Q - Q^*\right)$ for every $\epsilon \in (0,1)$, it follows from the previous lemma that Q_ϵ is also F^*-IC. Therefore, it must be that (Q_ϵ, F^*) is weakly worse for the monopolist than (Q^*, F^*) . In other words, we must have

$$\int \left(\pi_{Q_\epsilon}(\theta) - \pi_Q(\theta)\right) F^*(\mathrm{d}\theta) \leq 0$$

for all ϵ. Dividing this inequality by $\epsilon > 0$, and taking the limit as $\epsilon \searrow 0$, gives

$$\begin{aligned} 0 &\geq \frac{1}{\epsilon}\int \left(\pi_{Q_\epsilon}(\theta) - \pi_Q(\theta)\right) F^*(\mathrm{d}\theta) \\ &= \int \theta\left(Q - Q^*\right)(\theta) - \left(V_Q - V_{Q^*}\right)(\theta)F^*(\mathrm{d}\theta) \\ &\quad - \int \frac{1}{\epsilon}\left(\kappa\left(Q^*(\theta) + \epsilon\left(Q - Q^*\right)(\theta)\right) - \kappa\left(Q^*(\theta)\right)\right) F^*(\mathrm{d}\theta) \\ &\to \int \theta\left(Q - Q^*\right)(\theta) - \left(V_Q - V_{Q^*}\right)(\theta)F^*(\mathrm{d}\theta) \\ &\quad - \int \kappa'\left(Q^*(\theta)\right)\left(Q - Q^*\right)(\theta)F^*(\mathrm{d}\theta),\end{aligned}$$

where convergence follows from Beppo Levi's Theorem (e.g., Aliprantis and Border (2006) Theorem 11.18).[25] The lemma follows. □

We now work towards the main result of this section. This result establishes conditions under-

[25]Because κ is convex, the function $\epsilon \mapsto \frac{1}{\epsilon}\left(\kappa\left(q + \epsilon\left(\tilde{q} - q\right)\right) - \kappa\left(q\right)\right)$ is decreasing in ϵ for all $\tilde{q}$ and q.

which the following inequalities hold:

$$\int_{\theta\geq\theta^*} \kappa'\left(Q(\theta)\right) F\left(\mathrm{d}\theta\right) \geq \theta^*(1-F_-(\theta^*)), \tag{12}$$

$$\int_{\theta\geq\theta^*} \kappa'\left(Q(\theta)\right) F\left(\mathrm{d}\theta\right) \leq \theta^*(1-F_-(\theta^*)). \tag{13}$$

These inequalities compare the average marginal cost of the allocation provided to every θ above θ^* to θ^*. Mussa and Rosen (1978) provide conditions under which θ^* is larger or smaller than this average marginal production costs when information is exogenous.

Before discussing these inequalities for the endogenous information case, we first show that the following inequality implies (13):

$$\int_{\theta>\theta^*} \kappa'\circ Q(\theta) F\left(\mathrm{d}\theta\right) \leq \left(1-F\left(\theta^*\right)\right)\theta^*. \tag{14}$$

This inequality is the same as (13), except that the integral on the left hand side excludes θ^*, and the right hand side has $1-F(\theta^*)$ instead of $1-F_-(\theta^*)$.

Lemma 7. *If* (14) *holds for* $\theta^*\in[\underline{\theta}_F,\bar{\theta}_F)$*, then* (13) *also holds*

Proof. We first argue (14) implies $\kappa'\circ Q\left(\theta^*\right)<\theta^*$. To do so, assume (14) holds, and suppose $\kappa'\circ Q\left(\theta^*\right)\geq\theta^*$ for a contradiction. Because Q is F-ICC, the allocation Q is strictly increasing on $\left[\underline{\theta}_F,\bar{\theta}_F\right]\supseteq\left[\theta^*,\bar{\theta}_F\right]$, and so $\kappa'\circ Q(\theta)>\kappa'\circ Q\left(\theta^*\right)$ for all $\theta>\theta^*$, because κ' is strictly increasing. Therefore,

$$\theta^*<\int \kappa'\circ Q(\theta)F\left(\mathrm{d}\theta|\theta\in(\theta^*,\bar{\theta}_F]\right)=\int \kappa'\circ Q(\theta)F\left(\mathrm{d}\theta|\theta>\theta^*\right)=\frac{\int_{\theta>\theta^*}\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)}{1-F\left(\theta^*\right)},$$

contradicting (14). Thus, we have shown that $\kappa'\circ Q(\theta^*)<\theta^*$. Using this inequality, we now show that (14) implies (13). Specifically, (13) follows from the inequality chain

$$\begin{aligned}\int_{\theta\geq\theta^*}\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)&=\int_{\theta>\theta^*}\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)+\left(F\left(\theta^*\right)-F_-\left(\theta^*\right)\right)\kappa'\circ Q\left(\theta^*\right)\\&\leq\left(1-F\left(\theta^*\right)\right)\theta^*+\left(F\left(\theta^*\right)-F_-\left(\theta^*\right)\right)\theta^*=\left(1-F_-\left(\theta^*\right)\right)\theta^*.\end{aligned}$$

The proof is now complete. $\square$

We now state Lemma 8, which is the main result of this section.

Lemma 8. *Suppose (Q, F) is monopolist optimal, and that Q is F-ICC. Fix any $\hat{\theta} \in \Theta$, and let*

$$\theta^* = \max\{\hat{\theta}, \underline{\theta}_Q\}.$$

Then:

(i) If $I_F(\hat{\theta}) = 0$ and $Q(\bar{\theta}_F) < \bar{q}$, the inequality (12) holds for θ^*.

(ii) If $\hat{\theta} = \underline{\theta}$ and $\underline{\theta}_F > \underline{\theta}_Q$, the inequality (13) holds for θ^*.

(iii) If p_Q strictly increases at $\hat{\theta}$ and $\hat{\theta} > \underline{\theta}_Q$, the inequality (13) holds for θ^*.

To understand our proof, revisiting Mussa and Rosen (1978) is helpful. To show (12), Mussa and Rosen (1978) consider the change in the monopolist's profit due to a slight increase in the quality given to all types weakly above θ^*. If one starts from an optimal allocation, the change in the monopolist's revenues—represented by equation (12)'s right hand side—must be below the change in the monopolist's production costs, which are given by the left hand side of (12). The opposite inequality (13) is derived in a similar fashion by noting the monopolist cannot benefit from slightly *reducing* the quality given to all types above θ^*—provided that the allocation Q is strictly increasing at θ^*. The reason for this caveat is that, if Q is constant around θ^*, reducing the quality given to θ^* would result in non-monotone allocation, which would violate the buyer's incentive constraint.

It turns out that, by moving p, one can apply Mussa and Rosen's (1978) perturbation arguments when reasoning about the solution to (6), subject to two caveats. First, one needs to take care to perturb p in a way that results in a new F-marginal price. In particular, one cannot increase p at F-pooling points, and one must make sure Q_p remains in $[0, \bar{q}]$ over F's support. Second, shifting p only affects the induced allocation Q_p for types at which $p + c'$ is in the interval $[0, \bar{q}]$. Thus, a marginal increase in p for all types above θ only changes the allocation for types above $\underline{\theta}_{Q_p}$. Taking these caveats into account and applying the ideas of Mussa and Rosen (1978) delivers the result.

Proof of Lemma 8-(i). Suppose (Q, F) is monopolist optimal, and that $Q(\bar{\theta}_F) < \bar{q}$. The following inequality holds for every θ^* such that $I_F(\theta^*) = 0$:

$$\int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(\mathrm{d}\theta) \geq (1 - F_-(\theta^*)) \max\{\theta^*, \underline{\theta}_Q\}. \tag{15}$$

Let p be the F-marginal price associated with the allocation Q. For any $\epsilon > 0$ such that $p(\bar{\theta}_F) + \epsilon < \bar{q}$, define

$$p_\epsilon(\theta) = \begin{cases} p(\theta) & \text{if } \theta < \hat{\theta}, \\ p(\theta) + \epsilon & \text{otherwise.} \end{cases}$$

Because $I_F(\hat{\theta}) = 0$ and p is an F-marginal price, p_ϵ is also an F-marginal price. Note that $Q^{p^\epsilon} \geq Q_p$ because $p_\epsilon \geq p$, meaning that $\underline{\theta}_{Q_{p^\epsilon}} \leq \underline{\theta}_Q \leq \underline{\theta}_F$. Therefore, every $\theta \geq \underline{\theta}_F$ has $Q^{p^\epsilon}(\theta) = Q_p(\theta) + \epsilon \mathbf{1}_{\theta \geq \hat{\theta}}$. Below we prove that

$$\lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta) \right] = \left[\theta - \max\{\hat{\theta}, \underline{\theta}_Q\} \right]_+ . \tag{16}$$

Using this convergence, we can appeal to Lemma 6 to get

$$\begin{aligned}
0 &\geq \frac{1}{\epsilon} \left\{ \int \left[(\theta - \kappa' \circ Q_p(\theta))(Q^{p^\epsilon}(\theta) - Q_p(\theta)) - (V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta)) \right] F(\mathrm{d}\theta) \right\} \\
&= \frac{1}{\epsilon} \left\{ \int \left[(\theta - \kappa' \circ Q_p(\theta)) \epsilon \mathbf{1}_{\theta \geq \hat{\theta}} - (V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta)) \right] F(\mathrm{d}\theta) \right\} \\
&= \int_{\theta \geq \hat{\theta}} (\theta - \kappa' \circ Q_p(\theta)) F(\mathrm{d}\theta) - \frac{1}{\epsilon} \int (V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta)) F(\mathrm{d}\theta) \\
&\to \int_{\theta \geq \hat{\theta}} (\theta - \kappa' \circ Q_p(\theta)) F(\mathrm{d}\theta) - \int \left[\theta - \max\{\hat{\theta}, \underline{\theta}_Q\} \right]_+ F(\mathrm{d}\theta) \\
&= \int_{\theta \geq \hat{\theta}} \left[\max\{\hat{\theta}, \underline{\theta}_Q\} - \kappa' \circ Q_p(\theta) \right] F(\mathrm{d}\theta),
\end{aligned}$$

where the convergence follows from Beppo Levi's Theorem (e.g., Aliprantis and Border, 2006, Theorem 11.18), and the last equality follows from observing that $\underline{\theta}_Q \leq \underline{\theta}_F$ implies that $\max\{\hat{\theta}, \underline{\theta}_Q\} > \underline{\theta}_F$ if and only if $\hat{\theta} > \underline{\theta}_F$. Rearranging the above inequality then delivers part (i) of the lemma.

Thus, to complete the proof of this part, arguing that (16) holds suffices. We divide our argument into two cases.

Case 1 Suppose $\hat{\theta} \geq \underline{\theta}_Q$. Direct computation reveals

$$V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta) = \int_{\hat{\theta}}^{\theta \vee \hat{\theta}} \epsilon \mathrm{d}\tilde{\theta} = \epsilon[\theta \vee \hat{\theta} - \hat{\theta}] = \epsilon[\theta - \hat{\theta}]_+ = \epsilon \left[\theta - \max\{\hat{\theta}, \underline{\theta}_p\} \right]_+ .$$

where

$$\underline{\theta}_p := \underline{\theta}_{Q_p}.$$

The equality (16) immediately follows.

Case 2 Suppose $\hat{\theta} < \underline{\theta}_p$. If $\theta < \underline{\theta}_{Q_{p^\epsilon}}$, then $\theta \leq \underline{\theta}_Q$, and so $V_{Q^{p^\epsilon}}(\theta) - V_{Q_p}(\theta) = 0 = [\theta - \max\{\hat{\theta}, \underline{\theta}_Q\}]_+$, as required. The same equality also holds if $\theta < \hat{\theta}$. Consider then a $\theta \geq \max\{\underline{\theta}_{Q_{p^\epsilon}}, \hat{\theta}\}$. Then,

$$\begin{aligned}
V_{Q_{p^\epsilon}}(\theta) - V_{Q_p}(\theta) &= \int_{\underline{\theta}_Q}^{\underline{\theta}_Q \vee \theta} \epsilon \mathrm{d}\tilde{\theta} + \int_{\underline{\theta}_{Q_{p^\epsilon}}}^{\underline{\theta}_Q \wedge \theta} \epsilon \mathbf{1}_{\tilde{\theta} \geq \hat{\theta}} \mathrm{d}\tilde{\theta} \\
&= \epsilon[\underline{\theta}_Q \vee \theta - \underline{\theta}_Q] + \epsilon \left[\underline{\theta}_Q \wedge \theta - \underline{\theta}_{Q_{p^\epsilon}} \vee \hat{\theta} \right].
\end{aligned}$$

Therefore,

$$\frac{1}{\epsilon}\left[V_{Q_{p^\epsilon}}(\theta)-V_{Q_p}(\theta)\right]=[\underline{\theta}_p\vee\theta-\underline{\theta}_p]+\left[\underline{\theta}_p\wedge\theta-\underline{\theta}_{p^\epsilon}\vee\hat{\theta}\right]$$
$$\xrightarrow{\epsilon\to 0}[\underline{\theta}_p\vee\theta-\underline{\theta}_p]=[\theta-\underline{\theta}_p]_+=\left[\theta-\max\{\hat{\theta},\underline{\theta}_Q\}\right]_+,$$

where convergence follows from Q_{p^ϵ} uniformly converging to Q, and the last equality from the current case's assumption that $\hat{\theta}<\underline{\theta}_Q$.

□

Proof of Lemma 8-(ii). Since $\underline{\theta}_F>\underline{\theta}_Q$, $Q(\underline{\theta}_F)>0$. For any $\varepsilon\in(0,Q(\underline{\theta}_F))$, let $\theta_\varepsilon=\inf\{\theta:Q(\theta)\geq\varepsilon\}$, and define $p_\varepsilon(\cdot):=p(\cdot)-\varepsilon$. Observe p_ε is an F-marginal price because p is an F-marginal price, and $\epsilon<Q(\underline{\theta}_F)=p(\underline{\theta}_F)+c'(\underline{\theta}_F)$, meaning $p_\varepsilon(\underline{\theta}_F)>-c'(\underline{\theta}_F)$ (which is sufficient for $p_\varepsilon\geq -c'(\underline{\theta}_F)$ to hold).

Let $Q_\varepsilon:=Q_{p_\varepsilon}$ be the F-ICC allocation induced by p_ε. Obviously, $\theta_\varepsilon\leq\underline{\theta}_F$, and

$$Q_\varepsilon(\theta)=p_Q(\theta)+c'(\theta)-\varepsilon=Q(\theta)-\epsilon$$

for all $\theta\in\left[\theta_\varepsilon,\bar{\theta}_F\right]$. Noting that for all $\theta\in[\underline{\theta}_F,\bar{\theta}_F]$,

$$V_{Q_\varepsilon}(\theta)-V_Q(\theta)=\int_{\theta_\varepsilon}^{\theta}-\varepsilon\,\mathrm{d}\tilde{\theta}+\int_{\underline{\theta}}^{\theta_\varepsilon}-Q\left(\tilde{\theta}\right)\mathrm{d}\tilde{\theta}=\varepsilon\left(\theta_\varepsilon-\theta\right)-V_Q\left(\theta_\varepsilon\right),$$

and so by Lemma 6, we have

$$\begin{aligned}0&\geq\int\left[\left(\theta-\kappa'\circ Q(\theta)\right)(-\varepsilon)-\varepsilon\left(\theta_\varepsilon-\theta\right)+V_Q\left(\theta_\varepsilon\right)\right]F\left(\mathrm{d}\theta\right)\\&=\int\left[\left(\kappa'\circ Q(\theta)-\theta_\varepsilon\right)\varepsilon+V_Q\left(\theta_\varepsilon\right)\right]F\left(\mathrm{d}\theta\right).\end{aligned}$$

This inequality, however, implies that

$$\varepsilon\int_{\theta\geq\underline{\theta}_Q}\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)=\varepsilon\int\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)\leq\int\left[\varepsilon\theta_\varepsilon-V_Q\left(\theta_\varepsilon\right)\right]F(\mathrm{d}\theta)\leq\varepsilon\theta_\varepsilon.$$

Dividing both sides by ϵ, taking $\epsilon\searrow 0$, and noting that $\theta_\epsilon\searrow\underline{\theta}_Q$ completes the proof. □

Proof of Lemma 8-(iii) The proof of this part proceeds in 3 cases:

1. p_Q has an upward jump at θ^*.

2. p_Q strictly increases immediately below θ^*.

3. p_Q strictly increases immediately above θ^*.

We begin with the case in which p_Q jumps at θ^*.

Case 1: $p_{Q-}(\theta^*) < p_{Q+}(\theta^*)$**.** Observe Q being F-ICC and $p_{Q-}(\theta^*) < p_{Q+}(\theta^*)$ means $I_F(\theta^*) = 0$, and so $\theta^* > \underline{\theta}_F$. For any $\varepsilon \in (0, p_{Q+}(\theta^*) - p_{Q-}(\theta^*))$, define

$$p_\varepsilon(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta < \theta^*, \\ p_Q(\theta^*) \wedge (p_{Q+}(\theta^*) - \varepsilon) & \text{if } \theta = \theta^*, \\ p_Q(\theta) - \varepsilon & \text{if } \theta > \theta^*. \end{cases}$$

It is easy to verify that p_ε is an F-marginal price. Let $Q_\varepsilon := Q_{p_\varepsilon}$ be the F-ICC mechanism associated with p_ε. It follows Q_ε is F-IC, and thus one can apply Lemma 6 to get the following inequality for every $\varepsilon \in (0, p_{Q+}(\theta^*) - p_{Q-}(\theta^*))$,

$$\begin{aligned} 0 &\geq \int_{\theta > \theta^*} [(\theta - \kappa' \circ Q(\theta))(-\varepsilon) - \varepsilon(\theta^* - \theta)] F(\mathrm{d}\theta) \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*))(p_\varepsilon(\theta^*) - p_Q(\theta^*)) \\ &= \int_{\theta > \theta^*} (\kappa' \circ Q(\theta) - \theta^*) \varepsilon F(\mathrm{d}\theta) \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*))(p_\varepsilon(\theta^*) - p_Q(\theta^*)). \end{aligned}$$

Rearranging gives

$$\begin{aligned} \int_{\theta > \theta^*} \kappa' \circ Q(\theta) F(\mathrm{d}\theta) &\leq (1 - F(\theta^*))\theta^* \\ &\quad + (F(\theta^*) - F_-(\theta^*))(\theta^* - \kappa' \circ Q(\theta^*)) \left(\frac{p_\varepsilon(\theta^*) - p_Q(\theta^*)}{\varepsilon} \right). \end{aligned} \tag{17}$$

We now distinguish between two cases. Suppose first $p_Q(\theta^*) < p_{Q+}(\theta^*)$. Then for all small enough $\varepsilon > 0$, $p_\varepsilon(\theta^*) - p_Q(\theta^*) = 0$, and so equation (17) is equivalent to (13).

Suppose then $p_Q(\theta^*) = p_{Q+}(\theta^*)$. Then $p_Q(\theta^*) - p_\varepsilon(\theta^*) = \varepsilon$. Substituting into (17) and rearranging gives (13), as desired. □

Case 2: $p_{Q-}(\theta^*) = p_{Q+}(\theta^*)$ ***and*** $p_{Q-}(\theta^*) > p_Q(\theta)$ ***for all*** $\theta < \theta^*$***:*** We begin by arguing that we can find a sequence $(\theta_n)_{n \in \mathbb{N}}$ in $\left[\underline{\theta}_F, \bar{\theta}_F\right]$ such that $\theta_n \nearrow \theta^*$, $I_F(\theta_n) = 0$ for all n, and $p_Q(\theta_n) < p_Q(\theta_{n+1})$ for all n. We then use this sequence to construct a sequence of allocations that keep F incentive compatible. This allocation sequence, combined with Lemma 6, delivers a sequence of first-order conditions whose limit delivers (13).

Let us find the sequence $(\theta_n)_{n\in\mathbb{N}}$. For every $\delta > 0$, p_Q is non-constant on $[\theta^* - \delta, \theta^*]$, because if it were, $p_{Q-}(\theta^*) = p_Q(\theta^* - \delta) < p_{Q-}(\theta^*)$. It follows we can find a sequence $\left\{\tilde{\theta}_n\right\}_{n\in\mathbb{N}}$ in $(\underline{\theta}_F, \theta^*)$ with $\tilde{\theta}_n \nearrow \theta^*$ such that $p_Q\left(\tilde{\theta}_n\right) < p_Q\left(\tilde{\theta}_{n+1}\right)$ for all n. It follows p_Q is non-constant on $\left[\tilde{\theta}_m, \tilde{\theta}_n\right]$ for any $m < n$, and so every $m < n$ admits some $\theta_{m,n} \in \left[\tilde{\theta}_m, \tilde{\theta}_n\right]$ for which $I_F(\theta_{m,n}) = 0$. Choosing $\theta_n := \theta_{2n,2n+1}$, we have $\theta_n \nearrow \theta^*$, and

$$p_Q(\theta_n) = p_Q(\theta_{2n,2n+1}) \leq p_Q\left(\tilde{\theta}_{2n+1}\right) < p_Q\left(\tilde{\theta}_{2n+2}\right) \leq p_Q\left(\theta_{2(n+1),2(n+1)+1}\right) = p_Q(\theta_{n+1}),$$

meaning $(\theta_n)_{n\in\mathbb{N}}$ is as desired.

We now construct an F-ICC allocation for every θ_n in the above sequence. For this purpose, let $\delta_n = p_Q(\theta^*) - p_Q(\theta_n) > 0$,

$$p_n(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta \leq \theta_n \\ p_Q(\theta) - \delta_n & \text{if } \theta \geq \theta^* \\ p_Q(\theta_n) & \text{if } \theta \in [\theta_n, \theta^*]. \end{cases}$$

Since p_Q is an F-marginal price, and because $I_F(\theta^*) = 0$ (because p_Q is not constant around θ^*), the function p_n is an F-marginal price for every n, and so the allocation $Q_n := Q_{p_n}$ is an F-ICC allocation. Thus, Q_n is F-ICC.

Next, we apply Lemma 6 to get a first-order condition indexed by n. For this purpose, observe that for all $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$,

$$V_{Q_n}(\theta) - V_Q(\theta) = \int_{\theta_n \wedge \theta}^{\theta^* \wedge \theta} \left(p_Q(\theta_n) - p_Q\left(\tilde{\theta}\right)\right) \mathrm{d}\tilde{\theta} - \delta_n (\theta - \theta^* \wedge \theta).$$

Therefore, Lemma 6 delivers the following inequality for all n,

$$\begin{aligned}
0 \geq & \int_{\theta \geq \theta^*} (\theta - \kappa' \circ Q(\theta)) (-\delta_n) F(\mathrm{d}\theta) + \int_{\theta \in [\theta_n, \theta^*)} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta_n) - p_Q(\theta)) F(\mathrm{d}\theta) \\
& - \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} \left(p_Q(\theta_n) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} F(\mathrm{d}\theta) - \int_{\theta \geq \theta^*} -\delta_n (\theta - \theta^*) F(\mathrm{d}\theta) \\
= & \int_{\theta \geq \theta^*} (\kappa' \circ Q(\theta) - \theta^*) \delta_n F(\mathrm{d}\theta) + \int_{\theta \in [\theta_n, \theta^*)} (\theta - \kappa' \circ Q(\theta)) (p_Q(\theta_n) - p_Q(\theta)) F(\mathrm{d}\theta) \\
& - \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} \left(p_Q(\theta_n) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} F(\mathrm{d}\theta).
\end{aligned}$$

Rearranging and noting that $p_Q(\theta_n) \leq p_Q(\theta)$ for all $\theta \geq \theta_n$ delivers

$$\begin{aligned}
\int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(\mathrm{d}\theta) &\leq \int_{\theta \geq \theta^*} \theta^* F(\mathrm{d}\theta) - \int_{\theta \in [\theta_n, \theta^*)} (\theta - \kappa' \circ Q(\theta)) \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) F(\mathrm{d}\theta) \\
&\quad + \int_{\theta \geq \theta_n} \int_{\theta_n}^{\theta^* \wedge \theta} \frac{1}{\delta_n} \left(p_Q(\theta_n) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} F(\mathrm{d}\theta) \\
&\leq \int_{\theta \geq \theta^*} \theta^* F(\mathrm{d}\theta) - \int_{\theta \in [\theta_n, \theta^*)} (\theta - \kappa' \circ Q(\theta)) \frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta)) F(\mathrm{d}\theta) \\
&\leq \int_{\theta \geq \theta^*} \theta^* F(\mathrm{d}\theta) + \int_{\theta \in [\theta_n, \theta^*)} |\theta - \kappa' \circ Q(\theta)| \left|\frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta))\right| F(\mathrm{d}\theta).
\end{aligned} \tag{18}$$

We now show taking the limit of equation (18) as $n \to \infty$ delivers equation (13). To do so, observe $|p_Q(\theta_n) - p_Q(\theta)| \leq \delta_n$ for all $\theta \in [\theta_n, \theta^*)$, and that $\theta - \kappa' \circ Q(\theta) \leq \bar{\theta}_F + \kappa' \circ Q\left(\bar{\theta}_F\right)$ for all $\theta \in [\theta_n, \theta^*)$. Therefore, an M exists such that $|\theta - \kappa' \circ Q(\theta)| \left|\frac{1}{\delta_n} (p_Q(\theta_n) - p_Q(\theta))\right| \leq M$ for all n. Substituting back into (18) and taking limit with n delivers

$$\int_{\theta \geq \theta^*} \kappa' \circ Q(\theta) F(\mathrm{d}\theta) \leq \int_{\theta \geq \theta^*} \theta^* F(\mathrm{d}\theta) + M(F_-(\theta^*) - F(\theta_n)) \to \int_{\theta \geq \theta^*} \theta^* F(\mathrm{d}\theta).$$

Hence (13) holds at θ^*. □

Case 3: $p_{Q-}(\theta^*) = p_{Q+}(\theta^*)$ ***and*** $p_{Q+}(\theta^*) < p_Q(\theta)$ ***holds for all*** $\theta > \theta^*$***.*** We begin by finding a sequence $(\theta_n)_{n \in \mathbb{N}}$ in $\left[\underline{\theta}_F, \bar{\theta}_F\right]$ such that $\theta_n \searrow \theta^*$, $I_F(\theta_n) = 0$ for all n, and $p_Q(\theta_n) > p_Q(\theta_{n+1})$ for all n. We then construct a corresponding sequence of allocations that keep F incentive compatible for the buyer. This allocation sequence, combined with Lemma 6, delivers a sequence of first-order conditions whose limit delivers (13).

Let us find the sequence $(\theta_n)_{n \in \mathbb{N}}$. Observe that for every $\delta > 0$, p_Q is non-constant on $[\theta^*, \theta^* + \delta]$, because if it were constant, $p_{Q+}(\theta^*) = p_Q(\theta^* + \delta) > p_{Q+}(\theta^*)$. It follows we can find a sequence $(\tilde{\theta}_n)_{n \in \mathbb{N}}$ in $\left(\theta^*, \bar{\theta}\right)$ with $\tilde{\theta}_n \searrow \theta^*$ such that $p_Q(\tilde{\theta}_n) > p_Q(\tilde{\theta}_{n+1})$ for all n. To define $(\theta_n)_{n \in \mathbb{N}}$, observe that p_Q is non-constant on $[\tilde{\theta}_m, \tilde{\theta}_n]$ for any $m < n$, and so every $m < n$ admits some $\theta_{m,n} \in \left[\tilde{\theta}_m, \tilde{\theta}_n\right]$ for which $I_F(\theta_{m,n}) = 0$. Choosing $\theta_n := \theta_{2n,2n+1}$, we have $\theta_n \searrow \theta^*$, and

$$p_Q(\theta_n) = p_Q(\theta_{2n,2n+1}) \geq p_Q\left(\tilde{\theta}_{2n+1}\right) > p_Q\left(\tilde{\theta}_{2n+2}\right) \geq p_Q\left(\theta_{2(n+1),2(n+1)+1}\right) = p_Q(\theta_{n+1}).$$

Finally, observe $I_F(\theta) = 0$ and $\theta < \bar{\theta}$ implies $\theta < \bar{\theta}_F$. Hence, because $(\theta_n)_{n \in \mathbb{N}}$ is a strictly decreasing sequence, it has at most one element weakly above $\bar{\theta}_F$, and so it is without loss to take $(\theta_n)_{n \in \mathbb{N}}$ to be strictly below $\bar{\theta}_F$, as desired.

We now construct an F-ICC mechanism for every θ_n in the above sequence. Let $\delta_n :=$

$p_Q(\theta_n) - p_Q(\theta^*) > 0$. Define

$$p_n(\theta) = \begin{cases} p_Q(\theta) & \text{if } \theta \le \theta^* \\ p_Q(\theta) - \delta_n & \text{if } \theta \ge \theta_n \\ p_Q(\theta^*) & \text{if } \theta \in [\theta^*, \theta_n], \end{cases}$$

Note $I_F(\theta^*) = 0$ because p is not constant around θ^*. Using this fact and the fact that p is an F-marginal price, it is straightforward to verify that p_n is an F-marginal price as well for every n. We let $Q_n := Q_{p_n}$ be the F-ICC allocation induced by p_n.

Our next goal is to apply Lemma 6 to get a first-order condition indexed by n. For this purpose, observe that for $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$,

$$V_{Q_n}(\theta) - V_Q(\theta) = \int_{\theta^* \wedge \theta}^{\theta_n \wedge \theta} \left(p_Q(\theta^*) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} - \delta_n (\theta - \theta_n \wedge \theta).$$

Therefore, Lemma 6 delivers the following inequality for all n:

$$\begin{aligned} 0 \ge & \int_{\theta \ge \theta_n} (\theta - \kappa' \circ Q(\theta))(-\delta_n) F(\mathrm{d}\theta) + \int_{\theta \in [\theta^*, \theta_n)} (\theta - \kappa' \circ Q(\theta))(p_Q(\theta^*) - p_Q(\theta)) F(\mathrm{d}\theta) \\ & - \int_{\theta \ge \theta^*} \int_{\theta^*}^{\theta_n \wedge \theta} \left(p_Q(\theta^*) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} F(\mathrm{d}\theta) - \int_{\theta \ge \theta_n} -\delta_n (\theta - \theta_n) F(\mathrm{d}\theta) \\ = & \int_{\theta \ge \theta_n} (\kappa' \circ Q(\theta) - \theta_n) \delta_n F(\mathrm{d}\theta) + \int_{\theta \in [\theta^*, \theta_n)} (\theta - \kappa' \circ Q(\theta))(p_Q(\theta^*) - p_Q(\theta)) F(\mathrm{d}\theta) \\ & - \int_{\theta \ge \theta^*} \int_{\theta^*}^{\theta_n \wedge \theta} \left(p_Q(\theta^*) - p_Q(\tilde{\theta})\right) \mathrm{d}\tilde{\theta} F(\mathrm{d}\theta). \end{aligned}$$

Dividing both sides by δ_n and noting that $p_Q(\theta^*) \le p_Q(\theta)$ for all $\theta \ge \theta^*$ delivers

$$\begin{aligned} 0 \ge & \int_{\theta \ge \theta_n} (\kappa' \circ Q(\theta) - \theta_n) F(\mathrm{d}\theta) - \int_{\theta \in [\theta^*, \theta_n)} (\kappa' \circ Q(\theta) - \theta) \left(\frac{p_Q(\theta^*) - p_Q(\theta)}{\delta_n}\right) F(\mathrm{d}\theta) \\ \ge & \int_{\theta \ge \theta_n} (\kappa' \circ Q(\theta) - \theta_n) F(\mathrm{d}\theta) - \int_{\theta \in [\theta^*, \theta_n)} |\kappa' \circ Q(\theta) - \theta| \left|\frac{p_Q(\theta^*) - p_Q(\theta)}{\delta_n}\right| F(\mathrm{d}\theta). \end{aligned} \tag{19}$$

We now show taking the limit of equation (19) as $n \to \infty$ delivers equation (13). To do so, observe first $\mathbf{1}_{[\theta_n, \infty)}(\theta)(\kappa' \circ Q(\theta) - \theta_n)$ converges pointwise to $\mathbf{1}_{(\theta^*, \infty)}(\theta)(\kappa' \circ Q(\theta) - \theta^*)$. Second, notice $|p_Q(\theta^*) - p_Q(\theta)| \le \delta_n$ for all $\theta \in [\theta_n, \theta^*)$, and that $\theta - \kappa' \circ Q(\theta) \le \bar{\theta}_F + \kappa' \circ Q\left(\bar{\theta}_F\right)$ for all $\theta \in [\theta_n, \theta^*)$. Therefore, an M exists such that

$$|\theta - \kappa' \circ Q(\theta)| \left|\frac{1}{\delta_n}(p_Q(\theta_n) - p_Q(\theta))\right| \le M$$

for all n. Substituting these facts back into (19) gives

$$\begin{aligned}0 &\geq \int_{\theta\geq\theta_n}\left(\kappa'\circ Q(\theta)-\theta_n\right)F\left(\mathrm{d}\theta\right)-\int_{\theta\in[\theta^*,\theta_n)}\left|\kappa'\circ Q(\theta)-\theta\right|\left|\frac{p_Q\left(\theta^*\right)-p_Q(\theta)}{\delta_n}\right|F\left(\mathrm{d}\theta\right)\\&\geq \int \mathbf{1}_{[\theta_n,\infty)}(\theta)\left(\kappa'\circ Q(\theta)-\theta_n\right)F\left(\mathrm{d}\theta\right)-M\left(F_-\left(\theta_n\right)-F\left(\theta^*\right)\right)\\&\to \int \mathbf{1}_{(\theta^*,\infty)}(\theta)\left(\kappa'\circ Q(\theta)-\theta^*\right)F\left(\mathrm{d}\theta\right)=\int_{\theta>\theta^*}\kappa'\circ Q(\theta)F\left(\mathrm{d}\theta\right)-\theta^*\left(1-F\left(\theta^*\right)\right),\end{aligned}$$

where convergence follows from right continuity of F and the Lebesgue dominated convergence theorem. Hence, we've shown (14) holds for θ^*. Lemma 7 then implies (13) also holds, as desired.

□

We have now completed the proof of Lemma 8.

A.5.2. Information Perturbations

In this section we discuss the consequences of applying a perturbation based approach for analyzing the information design program (7). Such perturbations must satisfy two broad restrictions. The first restriction is that the perturbation must result in a signal—that is, a mean-preserving contraction of F_0. This restriction is satisfied whenever the perturbation creates a mean-preserving contraction of the original distribution, or when the perturbation creates a (small) mean-preserving spread over a set of F-pooling types. The second restriction is that the original allocation must be an ICC allocation with respect to the perturbed signal. This requirement means one can only alter F in regions where p is constant.

Lemma 9 outlines the consequence of two perturbation satisfying the above-mentioned restrictions. The lemma's first part identifies situations in which one can conduct mean-preserving contractions in the buyer's signal. Since such contractions cannot be profitable for the seller, they imply π_Q must satisfy the convex inequality (21). The lemma's second part identifies situations where one can spread the buyer's signal in mean-preserving manner. Profit maximization then delivers that π_Q must satisfy the concave inequality (20).

Lemma 9. *Let (Q^*, F^*) be monopolist-optimal, and suppose Q^* is F^*-ICC. Suppose p_{Q^*} is constant over $(\theta_*, \theta^*) \subseteq \left[\underline{\theta}_Q, \bar{\theta}_Q\right]$, and $F(\theta^*) > F_-(\theta_*)$. Then,*

(i) For every $\theta_1, \theta_2 \in \operatorname{supp} F^*\left(\cdot|\theta \in [\theta_*, \theta^*]\right)$ and every $\alpha \in [0, 1]$,

$$\pi_{Q^*}\left(\alpha\theta_1+(1-\alpha)\,\theta_2\right)\leq\alpha\pi_{Q^*}\left(\theta_1\right)+(1-\alpha)\,\pi_{Q^*}\left(\theta_2\right) \tag{20}$$

(ii) If $I_F(\theta) > 0$ for all $\theta \in [\theta_1, \theta_2] \subseteq [\theta_*, \theta^*]$, then

$$\pi_{Q^*}(\alpha\theta_1 + (1-\alpha)\theta_2) \geq \alpha\pi_{Q^*}(\theta_1) + (1-\alpha)\pi_{Q^*}(\theta_2) \tag{21}$$

for all $\alpha \in [0,1]$ such that $\alpha\theta_1 + (1-\alpha)\theta_2 \in \operatorname{supp} F^*(\cdot|\theta \in [\theta_*, \theta^*])$.

The lemma's first part says that, for appropriately chosen θ_1 and θ_2, the monopolist cannot benefit from contracting the mass on θ_1 and θ_2 into $\alpha\theta_1 + (1-\alpha)\theta_2$. Similarly, the lemma's second condition says the monopolist cannot benefit from the spreading mass on $\alpha\theta_1 + (1-\alpha)\theta_2$ across θ_1 and θ_2. For a rough proof sketch, consider the lemma's part (i), and suppose that F^* has atoms at θ_1 and θ_2. As explained above, that p_{Q^*} is constant on $[\theta_*, \theta^*]$ means one can pool together some mass from θ_1 and θ_2 without violating the buyer's incentive constraints. It follows that such pooling cannot benefit the monopolist; that is, (20) must hold. To prove the result without atoms, we approximate θ_1 and θ_2 with a shrinking neighborhood. The intuition for part (ii) of the lemma is similar: if equation (21) did not hold, the monopolist would strictly benefit from having the buyer spread the mass he puts on (a small neighborhood around) $\alpha\theta_1 + (1-\alpha)\theta_2$ across θ_1 and θ_2, thereby violating optimality of F^*.

Before discussing the formal proof, observe first that both parts of the lemma trivially hold when $\theta_1 = \theta_2$ or when $\alpha \in \{0, 1\}$. Therefore, suppose (without loss of generality) that $\theta_1 < \theta_2$.

Broadly speaking, the formal proof of the lemma proceeds as follows. Using that p_{Q^*} is constant on $[\theta_*, \theta^*]$, we construct a family of informational deviations which are incentive compatible for the buyer and that are indexed by $\epsilon > 0$. As ϵ vanishes, the difference between these deviations and F^* converges to the difference between an atom at $\alpha\theta_1 + (1-\alpha)\theta_2$ and a split of that atom's mass between an atom on θ_1 and an atom on θ_2 for the first part, and vice-versa for the second part. Then, we show the desired inequality using optimality of (Q^*, F^*) and continuity of $\pi_{Q^*}|_{[\theta_*, \theta^*]}$ (where the latter is implied by continuity of c' and p_{Q^*} being constant over $[\theta_*, \theta^*]$).

We now proceed with the actual proof. As a preliminary step, let $G = F^*(\cdot|\theta \in [\theta_*, \theta^*])$, $H = F^*(\cdot|\theta \notin [\theta_*, \theta^*])$, and $\beta = F^*(\theta^*) - F^*_-(\theta_*)$, and observe $F^* = \beta G + (1-\beta) H$. In addition, notice that N_{Q^*} is affine on $[\theta_*, \theta^*]$, because for any $\theta \in [\theta_*, \theta^*] \subseteq \left[\underline{\theta}_{Q^*}, \bar{\theta}_{Q^*}\right]$,

$$\begin{aligned} N_{Q^*} = N_{Q^*}(\theta_*) + \int_{\theta_*}^{\theta} (Q^* - c')(\theta)\mathrm{d}\theta &= N_{Q^*}(\theta_*) + \int_{\theta_*}^{\theta} p_{Q^*}(\theta)\mathrm{d}\theta \\ &= N_{Q^*}(\theta_*) + p_{Q^*}(\theta_*)(\theta - \theta_*), \end{aligned}$$

where the last equality follows from p_{Q^*} being constant on $[\theta_*, \theta^*] \subseteq \left[\underline{\theta}_{Q^*}, \bar{\theta}_{Q^*}\right]$.

Proof of Part (i). We begin by constructing the above-mentioned class of informational deviations. Take any $\epsilon \in \left(0, \frac{1}{2}(\theta_2 - \theta_1)\right)$ (which implies $[\theta_1 - \epsilon, \theta_1 + \epsilon] \cap [\theta_2 - \epsilon, \theta_2 + \epsilon] = \varnothing$), and define the following objects:

$$\begin{aligned}
G_{0,\epsilon} &= G\left(\cdot|\theta \notin [\theta_1-\epsilon,\theta_1+\epsilon] \cup [\theta_2-\epsilon,\theta_2+\epsilon]\right),\\
G_{1,\epsilon} &= G\left(\cdot|\theta \in [\theta_1-\epsilon,\theta_1+\epsilon]\right),\\
G_{2,\epsilon} &= G\left(\cdot|\theta \in [\theta_2-\epsilon,\theta_2+\epsilon]\right),\\
\gamma_{1,\epsilon} &= G(\theta_1+\epsilon) - G_-(\theta_1-\epsilon),\\
\gamma_{2,\epsilon} &= G(\theta_2+\epsilon) - G_-(\theta_2-\epsilon),\\
\gamma_{0,\epsilon} &= 1-\gamma_{1,\epsilon}-\gamma_{2,\epsilon}.
\end{aligned}$$

Clearly, $G = \sum_{i=0}^{2} \gamma_{i,\epsilon} G_{i,\epsilon}$. Moreover, since $\theta_1, \theta_2 \in \operatorname{supp} G$, both $\gamma_{1,\epsilon}$ and $\gamma_{2,\epsilon}$ are strictly positive for all $\epsilon > 0$. For any $\epsilon \in \left(0, \frac{1}{2}(\theta_2-\theta_1)\right)$, define

$$\begin{aligned}
\theta_\epsilon &= \int \theta \,\mathrm{d}\left(\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon}\right),\\
\tilde{\gamma}_\epsilon &= \min\{\gamma_{1,\epsilon}, \gamma_{2,\epsilon}\} > 0, \text{ and}\\
G_\epsilon &= \gamma_{0,\epsilon} G_{0,\epsilon} + \tilde{\gamma}_\epsilon \mathbf{1}_{[\theta_\epsilon,\infty)} + (\gamma_{1,\epsilon} - \alpha\tilde{\gamma}_\epsilon) G_{1,\epsilon} + (\gamma_{2,\epsilon} - (1-\alpha)\tilde{\gamma}_\epsilon) G_{2,\epsilon}.
\end{aligned}$$

In words, G_ϵ alters G by pooling $\alpha\tilde{\gamma}_\epsilon$ mass from the ϵ-ball around θ_1 and $(1-\alpha)\tilde{\gamma}_\epsilon$ mass from the ϵ-ball around θ_2 and pooling them to create an $\tilde{\gamma}_\epsilon > 0$ mass on θ_ϵ; that is,

$$G_\epsilon - G = \tilde{\gamma}_\epsilon \left(\mathbf{1}_{[\theta_\epsilon,\infty)} - (\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon})\right).$$

With the above in hand, we can finally define our informational pertubation: specifically, take $F_\epsilon = \beta G_\epsilon + (1-\beta) H$.

Next, we argue $F_\epsilon \in \mathcal{I}$ and that Q^* is $F_\epsilon-$IC. For the first claim, observe that because $\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon} \succ \mathbf{1}_{[\theta_\epsilon,\infty)}$, G_ϵ is less informative than G, and so $F_\epsilon \preceq F^* \preceq F_0$. That $F_\epsilon \in \mathcal{I}$ follows from $\preceq$ being transitive. To see Q^* is F_ϵ-IC for all $\epsilon \in \left(0, \frac{1}{2}(\theta_2-\theta_1)\right)$, observe that

$$\begin{aligned}
\int N_{Q^*}(\theta)\,(F_\epsilon - F^*)\,(\mathrm{d}\theta) &= \beta \int N_{Q^*}(\theta)\,(G_\epsilon - G)\,(\mathrm{d}\theta)\\
&= \beta\tilde{\gamma}_\epsilon \int N_{Q^*}(\theta) \left[\mathbf{1}_{[\theta_\epsilon,\infty)} - (\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon})\right](\mathrm{d}\theta) = 0,
\end{aligned}$$

where the last equality follows from $\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon} \succ \mathbf{1}_{[\theta_\epsilon,\infty)}$, the support of $\alpha G_{1,\epsilon} + (1-\alpha)G_{2,\epsilon}$ being contained in $[\theta_*, \theta^*] \subseteq \left[\underline{\theta}_Q, \bar{\theta}_Q\right]$, and N_{Q^*} being affine on $[\theta_*, \theta^*]$.

Now, because (Q^*, F^*) is monopolist-optimal, that Q^* is $F_\epsilon-$IC all small $\epsilon > 0$ means that

$\int \pi_{Q^*}\mathrm{d}F_\epsilon \leq \int \pi_{Q^*}\mathrm{d}F$. Rearranging this inequality, dividing by $\beta\tilde{\gamma}_\epsilon$, and taking ϵ to zero delivers

$$\begin{aligned}
0 \leq \frac{1}{\beta\tilde{\gamma}_\epsilon}\int \pi_{Q^*}(\theta)(F^* - F_\epsilon)(\mathrm{d}\theta) &= \frac{1}{\tilde{\gamma}_\epsilon}\int \pi_{Q^*}(\theta)\left(G - G_\epsilon\right)(\mathrm{d}\theta) \\
&= \left[\alpha \int \pi_{Q^*}(\theta)G_{1,\epsilon}(\mathrm{d}\theta) + (1-\alpha)\int \pi_{Q^*}(\theta)G_{2,\epsilon}(\mathrm{d}\theta)\right] - \pi_{Q^*}(\theta_\epsilon) \\
&\to \left(\alpha\pi_{Q^*}(\theta_1) + (1-\alpha)\pi_{Q^*}(\theta_2)\right) - \pi_{Q^*}\left(\alpha\theta_1 + (1-\alpha)\theta_2\right),
\end{aligned}$$

where convergence follows from continuity of $\pi_{Q^*}|_{[\theta_*,\theta^*]}$, convergence of $G_{1,\epsilon}$ and $G_{2,\epsilon}$ to $\mathbf{1}_{[\theta_1,\infty)}$ and $\mathbf{1}_{[\theta_2,\infty)}$ respectively, and $\theta_\epsilon \to \alpha\theta_1 + (1-\alpha)\theta_2$.

Proof of Part (ii). Suppose now $[\theta_1,\theta_2] \subseteq [\theta_*,\theta^*]$ is such that $I_{F^*}(\theta') > 0$ holds for all $\theta' \in [\theta_1,\theta_2]$, and that $\alpha \in (0,1)$ is such that $\theta_\alpha := \alpha\theta_1 + (1-\alpha)\theta_2 \in \operatorname{supp} G$. We begin by defining the following family of deviations. For any strictly positive $\epsilon < \min\{\theta_* - \theta_1, \theta_2 - \theta^*\}$, define

$$\begin{aligned}
G_{0,\epsilon}(\cdot) &:= G(\cdot|\theta \notin [\theta_\alpha - \epsilon, \theta_\alpha + \epsilon]), \\
G_{1,\epsilon}(\cdot) &:= G(\cdot|\theta \in [\theta_\alpha - \epsilon, \theta_\alpha + \epsilon]), \\
\theta_\epsilon &:= \int \theta G_{1,\epsilon}(\mathrm{d}\theta) \\
\gamma_\epsilon &:= G(\theta_\alpha + \epsilon) - G_-(\theta_\alpha - \epsilon).
\end{aligned}$$

Clearly, $G = (1-\gamma_\epsilon)G_{0,\epsilon} + \gamma_\epsilon G_{1,\epsilon}$. Observe $\gamma_\epsilon > 0$, because $\theta_\alpha \in \operatorname{supp} G$, and that an $\alpha_\epsilon \in (0,1)$ exists such that

$$\theta_\epsilon = \alpha_\epsilon\theta_1 + (1-\alpha_\epsilon)\theta_2,$$

by our choice of ϵ. Obviously, $\theta_\epsilon \to \theta_\alpha$, and $\alpha_\epsilon \to \alpha$. For a given $\tilde{\gamma} \in (0,\gamma_\epsilon)$, define

$$G_{\tilde{\gamma},\epsilon} = (1-\gamma_\epsilon)G_{0,\epsilon} + (\gamma_\epsilon - \tilde{\gamma})G_{1,\epsilon} + \tilde{\gamma}\left(\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1-\alpha_\epsilon)\mathbf{1}_{[\theta_2,\infty)}\right).$$

Clearly, $G_{\tilde{\gamma},\epsilon}$ is a CDF.

We now construct our informational deviation: set $F_{\tilde{\gamma},\epsilon} := \beta G_{\tilde{\gamma},\epsilon} + (1-\beta)H$ for all ϵ and $\tilde{\gamma}$ satisfying the above conditions. We begin by arguing that this deviation is a signal—that is, $F_{\tilde{\gamma},\epsilon} \in \mathcal{I}$—whenever $\tilde{\gamma}$ is sufficiently small (holding ϵ fixed). To do so, observe that the function $F \mapsto I_F(\theta)$ is affine for all θ, meaning that

$$I_{F^*} - I_{F_{\tilde{\gamma},\epsilon}} = \tilde{\gamma}\beta\left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}} + (1-\alpha_\epsilon)I_{\mathbf{1}_{[\theta_2,\infty)}} - I_{G_{1,\epsilon}}\right) < 0, \tag{22}$$

where the inequality follows from $G_{1,\epsilon} \prec \alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1-\alpha_\epsilon)\mathbf{1}_{[\theta_2,\infty)}$. Since the support of $G_{1,\epsilon}$, $\mathbf{1}_{[\theta_1,\infty)}$, and $\mathbf{1}_{[\theta_2,\infty)}$ is contained in $[\theta_1,\infty)$, it follows $I_{F_{\tilde{\gamma},\epsilon}}(\theta) = I_{F^*}(\theta) \geq 0$ for all $\theta \leq \theta_1$. Next, observe that for any $F \in \mathcal{F}$ and any $\theta \geq \max(\operatorname{supp} F)$, $\int_{\theta' \leq \theta} F(\theta')\, \mathrm{d}\theta' = \theta - \int \theta' \mathrm{d}F(\theta')$, and $I_{G_{1,\epsilon}}(\theta) = \alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}}(\theta) - (1-\alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}}(\theta)$ for all $\theta \geq \theta_2$, meaning that $I_{F_{\tilde{\gamma},\epsilon}}(\theta) = I_{F^*}(\theta) \geq 0$ holds for all such θ. Consider now the case $\theta \in (\theta_1, \theta_2)$. That I_F is continuous for all F, combined with I_{F^*} being strictly positive over $[\theta_1, \theta_2]$, implies a $\zeta := \min I_{F^*}([\theta_1, \theta_2]) > 0$ and that

$$\xi_\epsilon := \min_{\theta \in [\theta_1, \theta_2]} \left(\alpha_\epsilon I_{\mathbf{1}_{[\theta_1,\infty)}} + (1-\alpha_\epsilon) I_{\mathbf{1}_{[\theta_2,\infty)}} - I_{G_{1,\epsilon}} \right) > -\infty.$$

Recalling that $\xi_\epsilon \leq 0$ (due to (22)), one can see that whenever $\tilde{\gamma} < -\zeta/\beta\xi_\epsilon$, $\theta \in [\theta_1, \theta_2]$ implies

$$I_{F_{\tilde{\gamma},\epsilon}}(\theta) \geq I_{F^*}(\theta) + \tilde{\gamma}\beta\xi_\epsilon \geq \zeta + \tilde{\gamma}\beta\xi_\epsilon \geq 0.$$

Thus, we have shown $F_{\tilde{\gamma},\epsilon} \in \mathcal{I}$ for all $\tilde{\gamma} < -\zeta/\beta\xi_\epsilon$.

We now argue Q^* is $F_{\tilde{\gamma},\epsilon}$-IC for all above-mentioned ϵ and all $\tilde{\gamma} < -\zeta/\beta\xi_\epsilon$. To see this, observe that

$$\begin{aligned} \int N_{Q^*}(\theta) \left(F_{\tilde{\gamma},\epsilon} - F^*\right)(\mathrm{d}\theta) &= \tilde{\gamma}\beta \int N_{Q^*}(\theta) \left(\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1-\alpha_\epsilon)\mathbf{1}_{[\theta_2,\infty)} - G_{1,\epsilon} \right)(\mathrm{d}\theta) \\ &= \tilde{\gamma}\beta \left(\alpha_\epsilon N_{Q^*}(\theta_1) + (1-\alpha_\epsilon) N_{Q^*}(\theta_2) - N_{Q^*}(\theta_\epsilon) \right) = 0, \end{aligned}$$

where the last equality follows from $\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1-\alpha_\epsilon)\mathbf{1}_{[\theta_2,\infty)} \succeq G_{1,\epsilon}$, the support of $\alpha_\epsilon \mathbf{1}_{[\theta_1,\infty)} + (1-\alpha_\epsilon)\mathbf{1}_{[\theta_2,\infty)}$ and $G_{1,\epsilon}$ being contained in $[\theta_*, \theta^*]$, and N_{Q^*} being affine on $[\theta_*, \theta^*]$.

For the proof's last step, observe that because Q^* is $F_{\tilde{\gamma},\epsilon}$-IC for the buyer for all small ϵ and $\tilde{\gamma}$, monopolist optimality of (Q^*, F^*) implies

$$\begin{aligned} 0 \geq \frac{1}{\tilde{\gamma}} \int \pi_{Q^*}(\theta) \left(F_{\tilde{\gamma},\epsilon} - F^*\right)(\mathrm{d}\theta) &= \alpha_\epsilon \pi_{Q^*}(\theta_1) + (1-\alpha_\epsilon)\pi_{Q^*}(\theta_2) - \pi(\theta_\epsilon) \\ &\xrightarrow{\epsilon \to 0} \alpha \pi_{Q^*}(\theta_1) + (1-\alpha)\pi_{Q^*}(\theta_2) - \pi(\theta_\alpha), \end{aligned}$$

where convergence follows from $\theta_\epsilon \to \theta_\alpha$, $\alpha_\epsilon \to \alpha$, and π_{Q^*} being continuous on $[\theta_*, \theta^*]$. The desired inequality follows.

A.5.3. Proof of Theorem 3

Without loss, we can assume (Q, F) is a monopolist optimal outcome with the property that Q is F-ICC.

We begin by arguing that when κ is strictly convex, if (13) holds at some $\theta^* \in [\underline{\theta}_F, \bar{\theta}_F)$, the monopolist must be providing a buyer whose signal realization is θ^* with a product of inefficiently low quality. For an explanation, note that because the allocation Q is F-ICC, the allocation is strictly increasing over $[\underline{\theta}_F, \bar{\theta}_F)$. When κ is strictly convex, the marginal cost for quality provision is strictly increasing as well, and so equation (13) implies that

$$\kappa'\left(Q\left(\theta^*\right)\right) < \int_{\theta \geq \theta^*} \kappa'\left(Q(\theta)\right) F^*\left(\mathrm{d}\theta | \theta \geq \theta^*\right) \leq \theta^*;$$

that is, $Q\left(\theta^*\right)$ lies strictly below its efficient level.

Next, we argue the inequality (13) holds for every $\theta \in [\underline{\theta}_F, \bar{\theta}_F)$ at which p_Q is strictly increasing, and so $Q(\theta)$ lies below its efficient level. If $\theta > \underline{\theta}_F$, this claim directly follows from Lemma 8-(iii). For $\theta = \underline{\theta}_F$, distinguish two cases: either $\underline{\theta}_F > \underline{\theta}_Q$ and so we can apply Lemma 8-(ii), or $\underline{\theta}_F = \underline{\theta}_Q$, in which case $\kappa' \circ Q(\underline{\theta}_Q) = \kappa'(0) < \underline{\theta} \leq \underline{\theta}_F$. Either way, the inequality (13) applies. It follows quality is inefficiently low (strictly) at all such θ.

Next, we argue that quality is inefficiently low for any $\theta^* \in [\underline{\theta}_F, \bar{\theta})$ around which p_Q is constant. Thus suppose $\theta^* \in \operatorname{supp} F$ is such that p_Q is constant on $[\theta^* - \delta, \theta^* + \delta]$ for some $\delta > 0$. Our goal is to show $\kappa' \circ Q\left(\theta^*\right) < \theta^*$.

Let $\theta_* = \inf\left\{\theta \geq \underline{\theta}_F : p_{Q+}(\theta) = p_Q\left(\theta^*\right)\right\}$. We now argue (13) holds at θ_*. There are three cases to consider: $\theta_* = \underline{\theta}_F = \underline{\theta}_Q$, $\theta_* = \underline{\theta}_F > \underline{\theta}_Q$, and $\theta_* > \underline{\theta}_F$. In the first case we have $Q(\theta_*) = 0$ and so $\kappa' \circ Q(\theta_*) < \underline{\theta} \leq \underline{\theta}_F$. In the second case, the desired inequality follows from Lemma 8-(ii). In the third case, p_Q must be strictly increasing at θ_*, and so (13) must hold by Lemma 8-(iii), and so $\kappa' \circ Q(\theta_*) < \theta_*$.

Define $\bar{\theta}^* = (\theta^* + \delta) \wedge \bar{\theta}_F$, and let $G := F\left(\cdot | \theta \in \left[\theta_*, \bar{\theta}^*\right]\right)$. We claim (13) holds for

$$\theta' := \min\left(\operatorname{supp} G\right).$$

Clearly, we are done if $\theta' = \theta_*$. If $\theta' > \theta_*$, then $\theta_* \notin \operatorname{supp} G$, and so $F_-\left(\theta_*\right) = F\left(\theta_*\right) = F_-\left(\theta'\right)$. We therefore have the following inequality chain:

$$\begin{aligned}\left(1 - F_-\left(\theta'\right)\right)\theta' &> \left(1 - F_-\left(\theta'\right)\right)\theta_* = \left(1 - F_-\left(\theta_*\right)\right)\theta_* \\ &\geq \int_{\theta \geq \theta_*} \kappa' \circ Q(\theta) F\left(\mathrm{d}\theta\right) = \int_{\theta \geq \theta'} \kappa' \circ Q(\theta) F\left(\mathrm{d}\theta\right),\end{aligned}$$

where the weak inequality follows from (13) holding at θ_*. Thus, we have shown (14) holds at θ', and so (13) holds as well (see Lemma 7).

If $\theta^* = \theta'$, then (13) holds for θ^*, and so $\kappa' \circ Q\left(\theta^*\right) < \theta^*$, as explained after Lemma 8. Hence, there is nothing left to prove in this case. Thus, hereafter, we suppose $\theta^* \neq \theta'$. Since $\theta^* \in \operatorname{supp} G$, we must have $\theta^* > \theta'$.

We now argue p_Q is constant on $\left[\theta', \bar{\theta}^*\right]$. To do so, notice $\kappa' \circ Q\left(\theta'\right) < \theta'$ implies $Q\left(\theta'\right) = Q_+\left(\theta'\right)$ since Q jumps towards efficiency. Hence,

$$p_Q\left(\theta'\right) = Q\left(\theta'\right) - c'\left(\theta'\right) = Q_+\left(\theta'\right) - c'\left(\theta'\right) = p_{Q+}\left(\theta'\right) = p_Q\left(\theta^*\right),$$

where the last equality follows from $\theta' \geq \theta_*$. It follows p_Q is constant on $[\theta', \theta^*] \cup [\theta^* - \delta, \bar{\theta}^*) = [\theta', \bar{\theta}^*)$. Recalling $\bar{\theta}^* = \min\left\{\bar{\theta}_F, \theta^* + \delta\right\}$, it follows $p_Q\left(\bar{\theta}^*\right) = p_Q\left(\theta^*\right)$. Thus, we have shown p_Q is constant on $\left[\theta', \bar{\theta}^*\right]$.

Consider now the line segment connecting $(\theta', \pi_Q(\theta'))$ with $(\theta^*, \pi_Q(\theta^*))$,

$$\begin{aligned} &\varphi : [\theta', \theta^*] \to \mathbb{R}, \\ &\qquad \theta \mapsto \pi_Q(\theta') + \left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right)(\theta - \theta'). \end{aligned}$$

We claim $\varphi(\theta) \geq \pi_Q(\theta)$ for all $\theta \in [\theta', \theta^*]$. Obviously, $\varphi(\theta) = \pi_Q(\theta)$ whenever $\theta \in \{\theta', \theta^*\}$. For $\theta \in (\theta', \theta^*)$, we get the following inequality:

$$\begin{aligned} \varphi(\theta) &= \left(\frac{\theta^* - \theta}{\theta^* - \theta'}\right)\varphi(\theta') + \left(\frac{\theta - \theta'}{\theta^* - \theta'}\right)\varphi(\theta^*) \\ &= \left(\frac{\theta^* - \theta}{\theta^* - \theta'}\right)\pi_Q(\theta') + \left(\frac{\theta - \theta'}{\theta^* - \theta'}\right)\pi_Q(\theta^*) \geq \pi_Q(\theta), \end{aligned}$$

where the inequality follows from Lemma 9 part (i), which applies because p_Q is constant on $\left[\theta', \bar{\theta}^*\right]$.

Next, we show $\left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right)$ is strictly positive. For this purpose, fix any $\epsilon \in (0, \theta^* - \theta')$. Observe p_Q is constant on $\left[\theta', \bar{\theta}^*\right]$ means that

$$Q(\theta' + \epsilon) - Q_+(\theta') = Q(\theta' + \epsilon) - Q(\theta') = c'(\theta' + \epsilon) - c'(\theta').$$

It follows $Q'_+(\theta') = c''(\theta')$, delivering the following inequality chain,

$$\begin{aligned} \left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right) &= \frac{1}{\epsilon}\left[\varphi(\theta' + \epsilon) - \varphi(\theta')\right] \\ &\geq \frac{1}{\epsilon}\left[\pi_Q(\theta' + \epsilon) - \pi_Q(\theta')\right] \\ &= \frac{1}{\epsilon}\theta'\left(Q(\theta' + \epsilon) - Q(\theta')\right) - \frac{1}{\epsilon}\left[\kappa \circ Q(\theta' + \epsilon) - \kappa \circ Q(\theta')\right] \\ &\quad + Q(\theta' + \epsilon) - \frac{1}{\epsilon}\left[V_Q(\theta' + \epsilon) - V_Q(\theta')\right] \\ &\to (\theta' - \kappa' \circ Q(\theta'))\, c''(\theta') > 0, \end{aligned}$$

where convergence follows from the chain rule and $V'_{Q+}(\theta') = Q_+(\theta')$, and the strict inequality from c being strictly convex.

We now turn to establishing $\kappa' \circ Q(\theta^*) < \theta^*$, thereby concluding the proof. Toward this goal, notice again that for any $\epsilon \in (0, \theta^* - \theta')$,

$$Q(\theta^*) - Q(\theta^* - \epsilon) = c'(\theta^*) - c'(\theta^* - \epsilon),$$

because p_Q is constant on $\left[\theta', \bar{\theta}^*\right]$. Therefore, $Q'_-(\theta^*) = c''(\theta')$. So, we obtain the following inequality chain:

$$\begin{aligned}
0 < \left(\frac{\pi_Q(\theta^*) - \pi_Q(\theta')}{\theta^* - \theta'}\right) &= \frac{1}{\epsilon}\left[\varphi(\theta^*) - \varphi(\theta^* - \epsilon)\right] \\
&\leq \frac{1}{\epsilon}\left[\pi_Q(\theta^*) - \pi_Q(\theta^* - \epsilon)\right] \\
&= \frac{1}{\epsilon}\theta^*\left(Q(\theta^*) - Q(\theta^* - \epsilon)\right) - \frac{1}{\epsilon}\left[\kappa \circ Q(\theta^*) - \kappa \circ Q(\theta^* - \epsilon)\right] \\
&\quad + Q(\theta^* - \epsilon) - \frac{1}{\epsilon}\left[V_Q(\theta^*) - V_Q(\theta^* - \epsilon)\right] \\
&\to \left(\theta^* - \kappa' \circ Q(\theta^*)\right) c''(\theta^*),
\end{aligned}$$

where convergence follows from the chain rule and $V'_{Q-}(\theta^*) = Q_-(\theta^*)$. Since $c''(\theta^*) > 0$, the above inequality implies $\kappa' \circ Q(\theta^*) < \theta^*$, as required.

To conclude the proof, it remains to show that quality is efficient at $\bar{\theta}_F$ whenever $\bar{\theta}_F = \bar{\theta}$. Suppose $\bar{\theta}_F = \bar{\theta}$. By the above, one can approach $\bar{\theta}_F$ from below with a sequence of types that receive inefficiently low quality, meaning $Q_-(\bar{\theta}_F)$ must be weakly below the efficient level. Since $Q_+(\bar{\theta}_F) = Q_+(\bar{\theta}) = \bar{q}$, and Q jumps towards efficiency, it must then be that $\bar{\theta}_F$ gets the efficient quality. The result follows.

A.6. Proof of Corollary 3

The corollary follows from Theorem 2 and Lemma 8. For an explanation, note first that Theorem 2 implies it is without loss for Q to be an F-ICC allocation. For such allocations, $Q(\theta)$ is strictly between 0 and $\bar{q}$ for all types in $(\underline{\theta}_Q, \bar{\theta}_Q)$. Since $\underline{\theta}_F$ must be above $\underline{\theta}_Q$ and $\bar{\theta}_F$ must be below $\bar{\theta}_Q$, we get that $Q(\theta)$ is interior whenever $\theta \in (\underline{\theta}_F, \bar{\theta}_F)$. To see why $Q(\bar{\theta}_F) = \bar{q}$, suppose otherwise for a contradiction. Then applying Lemma 8-(i) for $\hat{\theta} = \underline{\theta}$ to get that (12) must hold for $\theta^* = \underline{\theta}_Q$, which is equivalent to $\kappa_0 > \underline{\theta}_Q \geq \underline{\theta} > \kappa_0$, a contradiction. Intuitively, when $\kappa(q) = \kappa_0 q$ for $\kappa_0 < \underline{\theta}$, and $Q(\bar{\theta}) < \bar{q}$, the seller can raise the quality given to all types by some $\epsilon > 0$, while increasing the transfers all those types pay by $\underline{\theta}\epsilon$. Since $\underline{\theta} > \kappa_0$, such a perturbation increases the monopolist's profits. It follows such perturbations cannot be feasible; that is, $Q(\bar{\theta}_F) = \bar{q}$.

A.7. Proofs from Section 5

Note that the main text explanations are sufficient for proving Proposition 2 and Corollary 4. Here, we prove Proposition 1 and Lemma 2.

Proof of Proposition 1. Let $\mathcal{B} \subseteq \mathcal{I}$ be the set of bi-poolings. To prove the proposition, we suppose F_0 is continuous and has full support and Q is an ICC allocation, and show that the set of extreme points of $\mathcal{I}(Q)$ is $\mathcal{I}(Q) \cap \mathcal{B}$. Consequently, Bauer's maximum theorem then implies there is some bi-pooling that solves (7) whenever Q is an information-cost cancelling mechanism. The proposition then follows from Theorem 2.

We now argue that every extreme point of $\mathcal{I}(Q)$ is a bi-pooling. To do so, we argue that $\mathcal{I}(Q)$ is a face of $\mathcal{I}$—i.e., if $G = \alpha G_1 + (1-\alpha)G_2$ holds for $G \in \mathcal{I}(Q)$, $G_1, G_2 \in \mathcal{I}$, and $\alpha \in (0,1)$, then $G_1, G_2 \in \mathcal{I}(Q)$. It immediately follows that every extreme point of $\mathcal{I}(Q)$ must be an extreme point of $\mathcal{I}$, and so must be a bi-pooling.

Thus, suppose G can be written as a convex combination of two elements G_1, G_2 in $\mathcal{I}$. Without loss, we can assume $\frac{1}{2}(G_1 + G_2) = G$. To complete the proof, we show $G_1, G_2 \in \mathcal{I}(Q)$. For this purpose, we argue two properties hold. First, $\mathrm{supp}(G_1)$ and $\mathrm{supp}(G_2)$ are both contained in $[\underline{\theta}_Q, \bar{\theta}_Q]$. And second, $I_{G_1}(\theta) = I_{G_2}(\theta) = 0$ whenever p_Q is strictly increasing at θ. To prove the first property, note that $\mathrm{supp}(G_1) \cup \mathrm{supp}(G_2) = \mathrm{supp}(G) \subseteq [\underline{\theta}_Q, \bar{\theta}_Q]$. To prove the second property, observe that $I_G(\theta) = \frac{1}{2}(I_{G_1}(\theta) + I_{G_2}(\theta))$ for all θ. Hence, for any θ at which p_Q is strictly increasing. Then,

$$0 \leq I_{G_1}(\theta) = 2I_G(\theta) - I_{G_2}(\theta) = -I_{G_2}(\theta) \leq 0,$$

where the first and last inequalities follow from G_1 and G_2 both being in $\mathcal{I}$. It follows $I_{G_1}(\theta) = 0$ and $I_{G_2}(\theta) = 0$. The proof is now complete. □

Proof of Lemma 2. To simplify notation, let $\underline{\theta}_p := \underline{\theta}_{Q_p}$. Notice it is without loss to assume every F-marginal price p is constant on $[\underline{\theta}, \underline{\theta}_p]$: amending p so that it satisfies this property relaxes the monopolist's constraints without impacting the monopolist's objective. Also, if $\underline{\theta}_p = \underline{\theta}$, then $\lim_{\theta \to \underline{\theta}} c'(\theta) > -\infty$ (otherwise, p cannot be bounded), and we so it is without loss to require $p(\underline{\theta}_p) \geq -c'(\underline{\theta}_p)$ (this is because $Q_p(\underline{\theta}_p) > 0$ requires $p(\underline{\theta}_p) > c'(\underline{\theta}_p)$, and $Q_p(\underline{\theta}_p) = 0$ whenever $p(\underline{\theta}) = -c'(\underline{\theta}_p)$).

Suppose $\tilde{p}$ is an F-marginal price satisfying the above-mentioned normalizations that solves the program (6). Define the set

$$\tilde{\mathcal{P}} = \left\{ p \in \mathcal{P}(F) : \underline{\theta}_p = \underline{\theta}_{\tilde{p}} \text{ and } p(\underline{\theta}_{\tilde{p}}) = \tilde{p}(\underline{\theta}_{\tilde{p}}) \right\}.$$

Observe $\tilde{\mathcal{P}}$ is compact in point-wise convergence (by Helly's Theorem) and convex. Since $\tilde{p}$ solves program (6) and $\tilde{p} \in \tilde{\mathcal{P}}$,

$$\operatorname*{argmax}_{p\in\tilde{\mathcal{P}}}\left[\int \pi_{Q_p}(\theta)F(\mathrm{d}\theta)\right] \subseteq \operatorname*{argmax}_{p\in\mathcal{P}(F)}\left[\int \pi_{Q_p}(\theta)F(\mathrm{d}\theta)\right]$$

Hence, to prove the proposition, it suffices to show the set on the left hand side of the above inclusion contains a p^* with the desired property. To show this containment, we establish two facts. First, the function $p \mapsto \Pi(Q_p, F)$ is affine over $\tilde{\mathcal{P}}$. And second, the extreme points of $\tilde{\mathcal{P}}$ satisfy the desired properties.

We first argue $p \mapsto \Pi(Q_p, F)$ is affine over $\tilde{\mathcal{P}}$. To do so, it is sufficient to show

$$p \mapsto \pi_{Q_p}(\theta) := \theta\,[p(\theta) + c'(\theta)]_+ - V_{Q_p}(\theta)$$

is affine over the same set for every $\theta \in [\underline{\theta}_F, \bar{\theta}_F]$. Fix any $p_1, p_2 \in \tilde{\mathcal{P}}$ and $\alpha \in (0,1)$. Observe,

$$\begin{aligned}
V_{Q^{(1-\alpha)p_1+\alpha p_2}}(\theta) &= \int_{\underline{\theta}}^{\theta} \Big((1-\alpha)p_1(\tilde{\theta}) + \alpha p_2(\tilde{\theta}) + c'(\tilde{\theta})\Big)_+ \mathrm{d}\tilde{\theta} \\
&= \int_{\underline{\theta}_{\tilde{p}}}^{\theta} \Big((1-\alpha)p_1(\tilde{\theta}) + \alpha p_2(\tilde{\theta}) + c'(\tilde{\theta})\Big)\,\mathrm{d}\tilde{\theta} \\
&= (1-\alpha)V_{Q_{p_1}}(\theta) + \alpha V_{Q_{p_2}}(\theta).
\end{aligned}$$

That $p \mapsto \pi_{Q_p}(\theta)$ is affine over $\tilde{\mathcal{P}}$ then follows from noting that

$$p \mapsto [\theta(p(\theta) + c'(\theta)) - \kappa((p(\theta) + c'(\theta))]$$

is an affine function of $p(\theta)$.

Next, we argue that every extreme point p^* of $\tilde{\mathcal{P}}$ satisfies the requirements of the proposition. Towards this goal, let $\bar{\mathcal{P}}$ be the set of all increasing functions from Θ to $[\tilde{p}(\underline{\theta}_{\tilde{p}}), \bar{q} - c'(\bar{\theta}_F)]$. Standard arguments (e.g. Börgers, 2015, Lemma 2.7) show the extreme points of $\bar{\mathcal{P}}$ consist of all elements of $\bar{\mathcal{P}}$ that strictly increase in at most one point, and take values in $\{\tilde{p}(\underline{\theta}_{\tilde{p}}), \bar{q} - c'(\bar{\theta}_F)\}$. It follows that to prove the proposition, it suffices to show the set ext$(\tilde{\mathcal{P}})$ of $\tilde{\mathcal{P}}$'s extreme points is contained in ext$(\bar{\mathcal{P}})$, the extreme points of $\bar{\mathcal{P}}$.

Thus, we now complete the proof by arguing that ext$(\tilde{\mathcal{P}}) \subseteq$ ext$(\bar{\mathcal{P}})$. En route to this goal, we first show that $\tilde{\mathcal{P}} \subseteq \bar{\mathcal{P}}$. To see this inclusion, note every $p \in \tilde{\mathcal{P}}$ is increasing and has $p \leq \bar{q} - c'(\bar{\theta}_F)$ by virtue of being an F-marginal price. Moreover, since every $p \in \tilde{\mathcal{P}}$ is constant on $[\underline{\theta}, \underline{\theta}_p]$, we also have $p \geq \tilde{p}(\underline{\theta}_{\tilde{p}})$.

To complete our argument, we fix an arbitrary $p \in \tilde{\mathcal{P}} \setminus$ ext$(\bar{\mathcal{P}})$, and show p is not in ext$(\tilde{\mathcal{P}})$. Because $p \notin$ ext$(\bar{\mathcal{P}})$, we can find distinct $p_1, p_2 \in \bar{\mathcal{P}}$ such that $p = 0.5(p_1 + p_2)$. We now argue

$p_1, p_2 \in \tilde{\mathcal{P}}$, which implies p cannot be an extreme point of $\tilde{\mathcal{P}}$. Observe first that $p_1, p_2 \in \mathcal{P}(F)$, because p being constant over an interval implies the same for both p_1, p_2. Second, note that $0.5(p_1 + p_2)(\underline{\theta}_{\tilde{p}}) = p(\underline{\theta}_{\tilde{p}}) = \tilde{p}(\underline{\theta}_{\tilde{p}}) \leq p_i$ for $i = 1, 2$, and so $p_i(\underline{\theta}_{\tilde{p}}) = \tilde{p}(\underline{\theta}_{\tilde{p}})$. Thus, it remains only to show that $\underline{\theta}_{p_i} = \underline{\theta}_{\tilde{p}}$. Suppose first $\underline{\theta}_{\tilde{p}} = \underline{\theta}$. In this case, $\tilde{p}(\underline{\theta}_{\tilde{p}}) = \tilde{p}(\underline{\theta}) \geq -c'(\underline{\theta})$, and so $p_i(\theta) + c'(\theta) \geq \tilde{p}(\underline{\theta}_{\tilde{p}}) + c'(\underline{\theta}) \geq 0$ for all θ and $i = 1, 2$, meaning $\underline{\theta}_{p_i} = \underline{\theta}$. Suppose now $\underline{\theta}_{\tilde{p}} > \underline{\theta}$. Since $p(\underline{\theta}_F) \geq -c(\underline{\theta}_F)$, we have $\underline{\theta}_F \geq \underline{\theta}_{\tilde{p}}$. It follows $\underline{\theta}_F > \underline{\theta}$, and so $I_F(\underline{\theta}_F) > 0$. Since I_F is continuous, some $\epsilon > 0$ exists such that $I_F(\theta) > 0$ for all $\theta \in (\underline{\theta}, \underline{\theta}_F + \epsilon)$. Consequently, $p(\theta) = \tilde{p}(\underline{\theta}_{\tilde{p}})$ must hold for all $\theta \in [\underline{\theta}, \underline{\theta}_F + \epsilon)$. It follows that for such θ, we must have $p_i(\theta) = p(\theta)$ as well. It follows that $\underline{\theta}_{p_i} = \underline{\theta}_p = \underline{\theta}_{\tilde{p}}$. □

Proof of Proposition 2. Once we have Lemma 2, the proof follows the same lines as in the main text. □

A.8. Extending Theorem 3 to Allow Efficient Exclusion

In this section we explain the argument that extends Theorem 3 to the case where $\kappa'(0) \geq \underline{\theta}$. Note that the argument establishing that $\bar{\theta} = \kappa'(Q(\bar{\theta}))$ whenever $\bar{\theta} \in \operatorname{supp}(F)$ easily extends to this case. Thus, all we need to do is argue that any $\theta \in \operatorname{supp}(F)$ such that $Q(\theta) > 0$ and $\theta < \bar{\theta}$, $\theta > \kappa'(Q(\theta))$. For brevity, we provide the major steps only, and skip some of the details. So, suppose (Q, F) is a monopolist optimal outcome in which Q is F-ICC. Our goal is to argue that $\theta \in \operatorname{supp}(F) \setminus \{\bar{\theta}\}$ has $\kappa'(Q(\theta)) \geq \theta$ only if $\theta = \underline{\theta}_Q$. Suppose then for a contradiction that another $\tilde{\theta} \in \operatorname{supp}(F) \setminus \{\underline{\theta}_Q, \bar{\theta}\}$ exists for which said inequality holds.

First, recall that (as shown in Section A.5.3) that π'_Q is positive for all $\theta \in \operatorname{supp}(F)$ on a given interval (θ_1, θ_2) where $I_F(\theta) > 0$ if $\kappa'(Q(\theta)) < \theta$, as by the envelope theorem,

$$\pi'_Q(\theta) = \theta Q'(\theta) - \kappa'(Q(\theta))Q'(\theta)$$

(recall that an F-ICC Q is differentiable whenever $I_F(\theta) > 0$). Conversely, if π' is negative, then $\kappa'(Q(\theta)) > \theta$. In addition, if p_Q is constant on some interval (θ_1, θ_2), then one can apply an argument similar to that in Theorem 3's proof to show that Lemma 9-(i) implies that $\pi'_Q(\theta) \leq \pi'_Q(\theta')$ for any $\theta < \theta'$ both in $\operatorname{supp}(F)$. Moreover, if I_F is strictly positive over (θ_1, θ_2), then a similar argument using Lemma 9-(ii) shows $\pi'_Q(\theta) \geq \pi'_Q(\theta')$, meaning $\pi'_Q(\theta) = \pi'_Q(\theta')$.

We now use the above to explain that $p_Q(\theta) = p_Q(\underline{\theta}_Q)$ for any $\theta \in \operatorname{supp}(F) \setminus \{\bar{\theta}\}$ at which $\kappa'(Q(\theta)) \geq \theta$. Fix any $\tilde{\theta} \in \operatorname{supp}(F)$ such that $\kappa'(Q(\tilde{\theta})) \geq \tilde{\theta}$. Suppose that $p_Q(\tilde{\theta}) > p_Q(\underline{\theta}_Q)$. By Lemma 8-(iii), p_Q must be constant around $\tilde{\theta}$. Define

$$\theta' = \sup\{\theta \in \Theta : \ p_Q \text{ strictly increases at } \theta, \ \theta \leq \tilde{\theta}\}.$$

Observe p_Q cannot be constant around θ'. Therefore, $\theta' < \tilde{\theta}$, and so $\kappa'(Q(\theta')) < \theta'$ by Lemma 8-(iii). But then one can apply the argument in Theorem 3's proof to show that $\kappa'(Q(\tilde{\theta})) < \tilde{\theta}$, a contradiction.

Next we establish that a θ^* exists with three properties. First, every $\theta \in \text{supp}(F)$ has $\kappa'(Q(\theta)) \geq \theta$ only if $\theta \leq \theta^*$, and $\kappa'(Q(\theta)) \leq \theta$ only if $\theta \geq \theta^*$. Second, $I_F(\theta^*) = 0$. And third, $p_{Q-}(\theta^*) = p(\underline{\theta}_Q)$. To do so, pick any $\tilde{\theta} \in \text{supp}(F) \setminus \{\underline{\theta}_Q\}$ such that $\kappa(Q(\tilde{\theta})) \geq \tilde{\theta}$. Let $\hat{\theta} = \sup\{\theta : p(\theta) = p(\underline{\theta}_Q)\}$. Note that $\hat{\theta} > \tilde{\theta}$. As explained earlier, π'_Q is increasing over $(\underline{\theta}_Q, \hat{\theta}) \cap \text{supp}(F)$. Since π'_Q has to be negative (positive) for quality to be above (below) efficient, we get that some $\theta^* \in [\underline{\theta}_Q, \hat{\theta}]$ exists such that for all $\theta \in \text{supp}(F) \cap (\underline{\theta}_Q, \hat{\theta})$, $\kappa'(Q(\theta)) \geq \theta$ only if $\theta \leq \theta^*$, and $\kappa'(Q(\theta)) \leq \theta$ only if $\theta \geq \theta^*$. We now explain that one can pick θ^* to satisfy $I_F(\theta^*) = 0$. First, suppose that $\pi'_Q(\theta)$ is constant at some value x for all $\theta \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$. Notice $x \leq 0$, because $\tilde{\theta} \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$ and $\kappa'(Q(\tilde{\theta})) \geq \tilde{\theta}$. In fact, in this case $\kappa'(Q(\theta)) \geq \theta$ for all $\theta \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$, and so we can take $\theta^* = \hat{\theta}$. To get that $I_F(\theta^*) = 0$ in this case, observe p_Q must be strictly increasing at $\hat{\theta}$ whenever $\hat{\theta} \neq \bar{\theta}$. This concludes the constant π'_Q case. For the second case, suppose we have $\theta, \theta' \in [\underline{\theta}_Q, \hat{\theta}] \cap \text{supp}(F)$ with $\theta < \theta'$ such that $\pi'_Q(\theta) < \pi'_Q(\theta')$, and $0 \in [\pi'_Q(\theta), \pi'_Q(\theta')]$. Note $\theta^* \in [\theta, \theta']$. Without loss, we can choose θ and θ' such that the sets $(\theta, \theta^*) \cap \text{supp}(F)$ and $(\theta^*, \theta') \cap \text{supp}(F)$ are both empty. Notice I_F cannot be strictly positive over $[\theta, \theta']$, because then one could apply the argument sketched in the second paragraph of this section to $(\theta - \epsilon, \theta' + \epsilon)$ for sufficiently small $\epsilon > 0$ to get that $\pi'_Q(\theta) = \pi'_Q(\theta')$. Thus, I_F must equal 0 at some $\theta'' \in [\theta, \theta']$, and we can choose $\theta^* = \theta''$.

We now claim that $\underline{\theta}_F = \underline{\theta}_Q$. If not, then since $Q(\bar{\theta}_F) < \bar{q}$, by Lemma 8-(ii),

$$\kappa'(Q(\underline{\theta}_F)) \leq \int \kappa'(Q(\theta)) dF(\theta) = \underline{\theta}_Q < \underline{\theta}_F$$

contradicting the claim that $\underline{\theta}_F \leq \kappa'(Q(\underline{\theta}_F))$.

We now show the monopolist can improve upon (Q, F) by looking at

$$\Theta_{F-} := \{\theta : \kappa'(Q(\theta)) > \theta\}$$

and replacing (Q, F) with $(\hat{Q}, \hat{F})$, where, letting

$$\theta_{F-} = \frac{\int_{\Theta_{F-}} \theta dF(\theta)}{F_-(\theta^*)}$$

we set

$$\hat{F}(\theta) = \begin{cases} 0, & \theta < \theta_{F-} \\ F_{-}(\theta^*), & \theta \in [\theta_{F-}, \theta^*) \\ F(\theta), & \theta > \theta^* \end{cases}$$

and

$$\hat{Q}(\theta) = \begin{cases} 0, & \theta \leq \theta_{F-} \\ c'(\theta) - c'(\theta_{F_-}), & \theta \in (\theta_{F_-}, \theta^*) \\ Q(\theta), & \theta \geq \theta^* \end{cases}$$

Notice that $\hat{Q}$ is a $\hat{F}$-ICC allocation for the marginal price function

$$\hat{p}(\theta) = \begin{cases} -c'(\theta_{F-}), & \theta < \theta^* \\ p_Q(\theta) + c'(\theta_{F-}) - c'(\underline{\theta}_F), & \theta \geq \theta^* \end{cases}$$

The new profit $\Pi_{\hat{Q}} \coloneqq \int \pi_{\hat{Q}}(\theta)\hat{F}(\mathrm{d}\theta)$ is now given by

$$\begin{aligned} \int \pi_{\hat{Q}}(\theta)\hat{F}(\mathrm{d}\theta) &= \int_{\theta<\theta^*} \pi_{\hat{Q}}(\theta)\hat{F}(\mathrm{d}\theta) + \int_{\theta\geq\theta^*} \pi_{\hat{Q}}(\theta)F(\mathrm{d}\theta) \\ &= \int_{\theta<\theta^*} (0)\hat{F}(\mathrm{d}\theta) + \int_{\theta\geq\theta^*} [\pi_Q(\theta) + V_Q(\theta_{F-})]F(\mathrm{d}\theta) \\ &> \int_{\theta<\theta^*} (0)F(\mathrm{d}\theta) + \int_{\theta\geq\theta^*} \pi_Q(\theta)F(\mathrm{d}\theta) \\ &\geq \int_{\theta<\theta^*} \pi_Q(\theta)F(\mathrm{d}\theta) + \int_{\theta\geq\theta^*} \pi_Q(\theta)F(\mathrm{d}\theta) \\ &= \int \pi_Q(\theta)F(\mathrm{d}\theta), \end{aligned}$$

where the weak inequality follows from noting that $\pi_Q(\underline{\theta}_Q) = 0$ and $\pi_Q^{'}(\theta) \leq 0$ for all $\theta \in [\underline{\theta}_Q, \theta^*)$. Thus, $(\hat{Q}, \hat{F})$ is an improvement over (Q, F). A contradiction.